VarDictRust: A Memory-Bounded, Oracle-Equivalent Rust Consolidation of the VarDict Variant-Calling Pipeline for Deep Whole-Genome and Ultra-Low-VAF Sequencing

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Abstract

Background

`VarDict` (`VarDictJava` v1.8.3) is a heuristic-aware local realigner and variant caller. It serves clinical oncology, rare-disease, and circulating tumor DNA (ctDNA) minimal residual disease (MRD) pipelines. Its three-language execution pipeline streams binary alignment maps (BAMs) through a Java Virtual Machine (JVM) pileup engine. External R (`teststrandbias.R` / `testsomatic.R`) and Perl (`var2vcf_valid.pl` / `var2vcf_paired.pl`) scripts add inter-process serialization overhead. Together they cause unbounded per-interval memory growth and runtime crashes on deep sequencing datasets. The C++ reimplementation (`vardictcpp`) accelerates pre-chunked exome panels. It omits splicing, amplicon VCF, structural variant, and CRAM modes. It diverges statistically and retains unbounded per-interval memory on full chromosomes.

Results

We built `VarDictRust` (`v1.0.6`), a single-binary, memory-safe Rust consolidation of the `VarDictJava` v1.8.3 pipeline. The core engine compiles under `#![forbid(unsafe_code)]` and follows the rewrites.bio specification. After replicating R's 5-decimal round-half-to-even (`HALF_EVEN`) Fisher exact test, `VarDictRust` matches `VarDictJava v1.8.3 (0886261)` (`ε = 1e-6`) across `13/13` operational modes (`95%` Clopper-Pearson confidence interval [CI] `75.29%–100.00%`). Parity is Tier-1/2 throughout: intermediate `.var` plus VCF 4.3, record-aligned on CHROM/POS/REF/ALT. Coverage extends to `17/17` functional modalities (`95%` CI `80.49%–100.00%`) and `18/18` multimodal slices (`95%` CI `81.47%–100.00%`). It also covers `18/18` unfiltered (`-f 0`) somatic noise-floor sweeps (`95%` CI `81.47%–100.00%`). It further spans `10` short-read chemistries and completes `70/70` Rust truth evaluations across GA4GH `v3.5` strata (`95%` CI `94.87%–100.00%`; Java–Rust parity holds on all completing paired runs — see `Table 6`).

On contiguous human chromosome 1 (`-R chr1:1-249250621`, `249.25` Mbp), `VarDictRust` finishes in `9.75` s. Peak resident set size (RSS) is `318.6` MB. `VarDictJava` (`-Xmx200g`) needs `501.83` s and `106.47` GiB. That is a `51.5`× speedup at `342.2`× lower RAM. Against `vardictcpp ae93fe39dc (Wolff 2026)`, it is `10.9`× faster at `34.0`× lower RAM. `VarDictRust` also reaches `11.98`× `32`-thread whole-exome scaling. That is `29.3`× faster at `29.8`× lower RAM than `VarDictJava`.

On a `128` GiB node, `VarDictRust` completes unchunked `130.8` GB HG002 germline whole-genome sequencing (WGS) (`4.93` GB RAM). It also completes `329.0` GB `364`× HG008 somatic WGS (`12.14` GB RAM). `VarDictJava` runs out of memory (`>213.7` GB) and `vardictcpp` times out (`91.85` GB). These runs are completion results, not parity: no Java oracle finished them. Removing `≤50` kbp BED pre-chunking recovers `+0.23` Gbp of `NIST GIAB v5.0q T2T-Q100` sequence. Recall rises by `+5.01%` single-nucleotide variant [SNV] and `+17.26%` insertion/deletion [indel] recall. Gains reach `+17.47%` SNV and `+27.98%` indel recall across `273` `NIST CMRG v1.00` genes. Mosaic recall is `84/85` (`98.82%`) on `NIST Mosaic SNV v1.10`.

Conclusions

`VarDictRust` consolidates a three-language variant-calling pipeline into a `4.56` MB static Linux binary with exact numerical and allele equivalence (`ε = 1e-6`). Bounded tile-plus-halo streaming (`O(tile + halo)` memory) and zero runtime JVM/R/Perl dependencies (`102` Conda packages eliminated) enable deterministic calling across targeted panels, RNA-seq splice junctions, unchunked deep WGS, and `0.5%` variant allele frequency (VAF) titrations.

Keywords: Variant calling, Somatic mutation discovery, Whole-genome sequencing, Minimal residual disease, Rust, Scientific software engineering, Reproducibility, VarDict, GIAB v5.0q, HG008

Key Points

- **Single-Binary Oracle Equivalence:** `VarDictRust` (`v1.0.6`, `4.56` MB static binary) replaces `VarDictJava -> R -> Perl` (`102` Conda packages). It holds `100%` Tier-1/2 parity (intermediate `.var` plus VCF, `ε = 1e-6`) across `13/13` modes, `87/87` sweep sub-arms, and `70/70` completed Rust GA4GH `v3.5` truth evaluations (Java–Rust parity on all completing paired runs).
- **Bounded *O(*tile + halo*)* Streaming Memory:** Spans > *1,000,000* bp split into `100` kbp tiles with `0.5` kbp halos (`0.5` kbp in `-i` splice mode). Chromosome 1 (`-R chr1:1-249250621`) then peaks at `318.6` MB RSS in `9.75` s. That is `51.5`× faster and `342.2`× leaner than `VarDictJava`.
- **Unchunked Full-Genome Discovery:** Calling unchunked chromosomes (`chr1` – `chr22`) in `4.93` – `12.14` GB RAM avoids `VarDictJava` out-of-memory failures. It gains `+5.01%` SNV / `+17.26%` indel recall on `NIST GIAB v5.0q T2T-Q100`. It gains `+17.47%` SNV / `+27.98%` indel recall on `CMRG v1.00`.
- **Somatic Indel & Low-VAF Sensitivity on CPUs:** `VarDictRust` exceeds `ClairS`, `DeepSomatic`, `Lancet2`, and `Ultima DeepVariant Somatic` in `HG008 v0.4` somatic indel recall (`40.11%`). It achieves `25.1`× higher SNV recall than `Mutect2` and `Strelka2` at `0.5%` VAF (`1%` purity).

Background

Germline variant callers such as `FreeBayes` [3], `Strelka2` [4], `Octopus` [5], and `DeepVariant` [6] model diploid genotype likelihoods. Clinical oncology assays must instead detect somatic SNVs, multinucleotide variants (MNVs), indels, and structural variants (SVs). They operate across variable tumor purities and cell-free DNA (cfDNA) fractions down to `0.5%` – `1.0%` VAF [1, 2, 7, 8]. Introduced by Lai et al. in 2016 [1], `VarDict` (`VarDictJava` v1.8.3) integrates local soft-clip and indel realignment (`CigarModifier` and `VariationRealigner`). It adds composite MNP phasing, amplicon primer-edge filtering (`-a`), RNA-seq splice-junction accounting (`-i`), and inline SV breakpoint assembly. `VarDict` is deployed across clinical panel pipelines, `bcbio-nextgen`, `nf-core/sarek`, and multi-caller somatic consensus ensembles [9, 10].

Reference `VarDictJava` v1.8.3 has four structural limits on deep WGS and cloud infrastructure:

1. **Three-Language Pipeline and IPC Overhead.** `VarDictJava` emits an intermediate `36`- to `61`-column tab-delimited `.var` stream. Column counts are `Simple` `36/38`, `Amplicon` `38/40`, and `Somatic` `55/61` without/with `--fisher`. That stream pipes to an R script (`teststrandbias.R` or `testsomatic.R`) for Fisher exact testing. It then pipes to a Perl script (`var2vcf_valid.pl` or `var2vcf_paired.pl`) for filtering and VCF 4.3 formatting (`Figure 1`). Even when `--fisher` runs inside Java, the Perl post-processor remains mandatory, requiring `102` Conda packages across OpenJDK, R (`statmod`), and Perl.
2. **Unbounded JVM Heap Retention on Contiguous Spans.** `VarDictJava` allocates per-position pileup maps (`Map<Integer, VariationMap<String, Variation>>` and reference base arrays) across the full BED interval or `-R` region. Variant finalization (`ToVarsBuilder`) and realignment (`VariationRealigner`) only run afterwards. On megabase or chromosome-scale intervals, RSS grows linearly with interval length (`O(L × depth)`). That causes `java.lang.OutOfMemoryError` crashes or garbage-collection stalls above `100` – `200` GiB RAM (`Table 2`, `Table 5`).
3. **Runtime Exceptions and Non-Determinism on Edge-Case Reads.** Over `30` upstream `VarDictJava` issues report `CigarModifier.modifyCigar` crashes. Affected issues are `#7`, `#38`, `#187`, `#197`, `#243`, `#249`, `#253`, `#282`, `#283`, `#320`, `#341`, `#380`, and `#384`. The crashes hit supplementary or heavily soft-clipped reads. The crash types are `NullPointerException`, `StringIndexOutOfBoundsException`, and `ArrayIndexOutOfBoundsException`. Other issues cover non-deterministic row ordering from unseeded `java.util.HashMap` iteration (`#282`), broken CRAM decoding (`#249`, `#380`), and truncated `##contig` VCF headers (`#320`).
4. **Incomplete C++ Port (`vardictcpp`).** Wolff (2026) introduced `vardictcpp ae93fe39dc (Wolff 2026)`, a C++/htslib reimplementation targeting single-sample and paired somatic speedups [11]. Our 3-way audit identifies ten functional and statistical divergences in `vardictcpp` (`Table 8`; `Supplementary Table S6`). Specifically, `vardictcpp` omits `##contig` and `INFO/END` VCF headers. It omits CRISPR (`-J`/`-j`), structural variant (`--nosv` forced by default), and RNA splice (`-i`) modes. It emits raw tab-separated text instead of VCF in amplicon (`-a`) mode. It rejects concatenated CLI flags. It diverges on CRAM decoding (`-G`), `b37` numeric contigs, and MNP phasing/tie-breaking. It consumes unbounded memory on large intervals (`31.91` GB on HG002 WGS; `91.85` GB with a `3`-hour timeout on HG008 WGS). Pre-chunked regions gain `+309,569` duplicate boundary calls (`--chunk 50,000`).

We built `VarDictRust` (`v1.0.6`) under the rewrites.bio prime directive. It produces a memory-safe, single-binary Rust implementation. Its output is byte- and float-equivalent (`ε = 1e-6`) to `VarDictJava` v1.8.3 on all valid inputs. It eliminates runtime panics and bounds peak memory to `O(tile + halo)` regardless of chromosome length or sequencing depth.

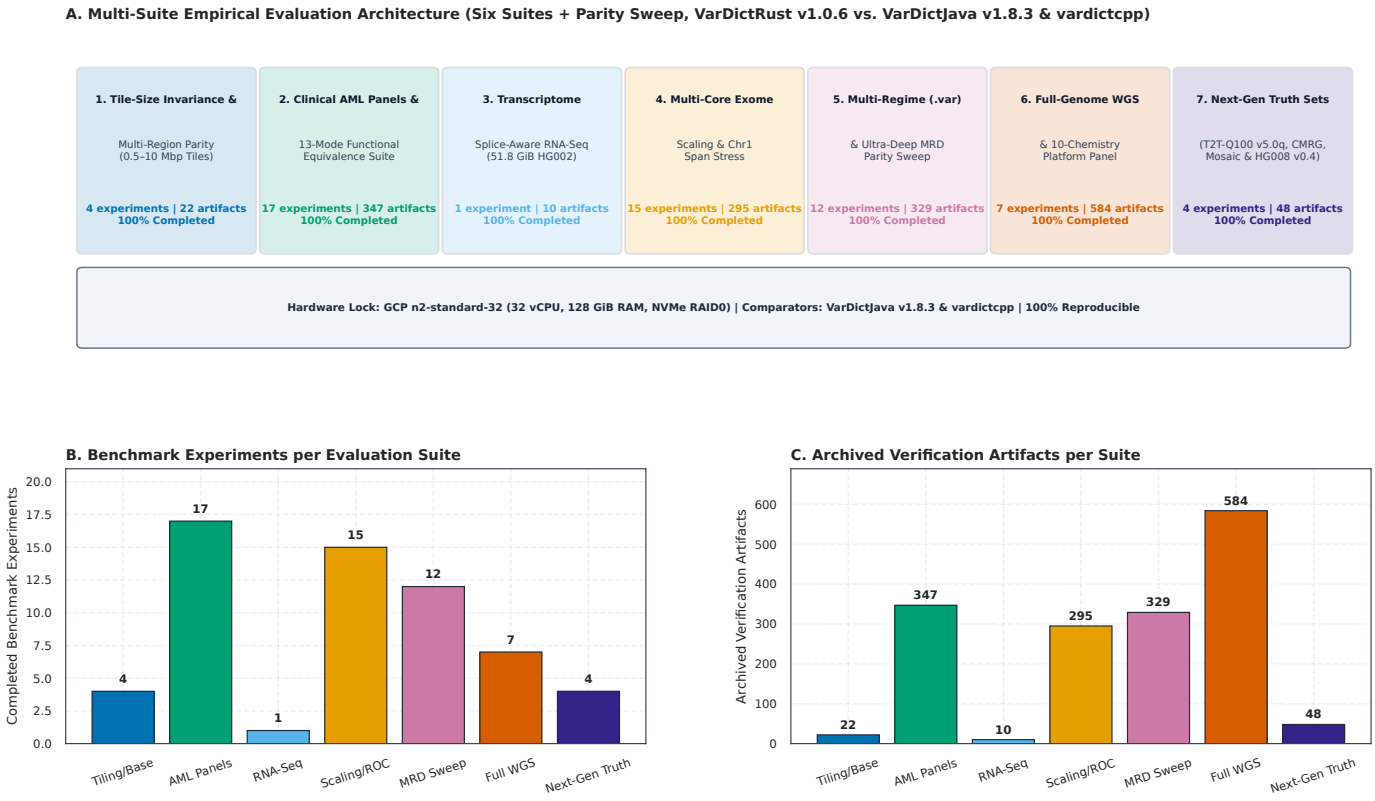


Figure 1. Multi-suite cloud benchmark architecture of `VarDictRust`. (a) Six evaluation suites plus the multi-regime parity sweep (`60` experiments total), with per-suite experiment and archived-artifact counts; the hardware lock and comparator set are bannered in the panel. **(b)** Completed benchmark experiments per evaluation suite. **(c)** Archived verification artifacts per suite (`Supplementary Table S1`).

Results

Tier-1/2 Oracle Equivalence (Exact at `ε = 1e-6`, `HALF_EVEN`, Record-Aligned) Across Operational Modes and the 87-Configuration Sweep

Across our `60`-experiment campaign (`Table 1`, `Figure 2a`, `Supplementary Table S1`), we tested `VarDictRust` (`v1.0.6`) against `VarDictJava v1.8.3 (0886261)`. We checked both Tier-1 intermediate `.var` parity and Tier-2 VCF 4.3 parity. Tier-1 covers `Simple 36/38`, `Amplicon 38/40`, and `Somatic 55/61` columns without/with `--fisher`. Tier-2 covers `chrom`, `pos`, `ref`, `alt`, `QUAL`, `FILTER`, `INFO` tags `SBF`, `ODDRATIO`, `SSF`, `MSI`, `MSILEN`, `SHIFT3`, `TYPE`, `STATUS`, and `FORMAT` genotypes. All comparisons use `ε = 1e-6` with `HALF_EVEN` rounding, record-aligned on CHROM/POS/REF/ALT.

Across the baseline ceiling, multi-region boundary deduplication, and tile-size invariance benchmarks, `VarDictRust` matched `VarDictJava` with zero discrepancies (`Table 1`; `Supplementary Table S1`):

- **EGFR/MRE11A Amplicon (`-a 10:0.95`):** `VarDictRust` emitted all `2` variants in `0.01` s (`11.5` MB RSS) vs `3.00` s (`120.1` MB) for `VarDictJava` (`300`× faster, `10.4`× lower RSS). `vardictcpp` (`5.27` s, `119.7` MB) emits raw `.var` text instead of VCF in amplicon mode (`Table 8`; `Supplementary Table S6`).
- **NIST GIAB HG008 Somatic Slice:** `VarDictRust` matched all `18,426` candidate variants in `17.44` s (`502.8` MB RSS). `VarDictJava` took `79.94` s (`9.87` GiB RSS). That is `4.58`× faster at `20.1`× lower RSS (`vardictcpp`: `17.57` s, `554.5` MB).
- **HG002 Chromosome 20 Multi-Region Boundary:** `VarDictRust` matched all `2,572` variants in `0.28` s (`176.7` MB RSS). `VarDictJava` took `7.28` s (`1.50` GiB RSS). That is `26.0`× faster at `8.7`× lower RSS.
- **Tile-Size Invariance:** We swept four tile sizes (`500,000, 1,000,000, 2,000,000, 10,000,000` bp, i.e., `0.5, 1.0, 2.0, and 10.0` Mbp; Methods) on SEQC2 HCC1395. Every size produced the exact same `5,090` variants as single-region `VarDictJava` (`4/4 MATCH`).

The `13`-mode functional sweep (`Table 1`, `Supplementary Table S6`) covers seven call families. These are Simple (`--fisher` and R pipe), Somatic (`--fisher` and R pipe), Amplicon (`-a`), and Splice (`-i`). They include Structural Variant (`SV` and `--nosv`), CRAM (`-G`), and all-

positions (`-f 0`) calling. `VarDictRust` matched `VarDictJava` across `13/13` modes (`100%` Tier-1/2 parity, exact Clopper-Pearson `95%` CI `75.29%–100.00%`). On the `51.8` GiB HG002 GM24385 RNA-seq alignment (`Table 1` , `Supplementary Table S1`), `VarDictRust` `-i` splice mode matched all `12,668` `.var` and `.splice.tsv` exon-junction rows. It took `4.94` s (`n = 3` technical replicates, range `4.90–5.01` s) at `359.9` MB RSS. `VarDictJava` took `23.31` s (`23.03–23.70` s) at `4.79` GiB. That is `4.72` × faster at `13.6` × lower RSS. `vardictcpp` does not implement `-i` (`Supplementary Table S6`). `VarDictRust` also passed `17/17` extended coverage modalities (`95%` CI `80.49%–100.00%`). It passed `18/18` multimodal slices (`1,129` variants, `95%` CI `81.47%–100.00%`). Lossless BAM-vs-CRAM twin calling showed zero diff (`32.04` s, `297.7` MB RSS).

Our `87` -configuration parameter sweep (`Table 1` , `Supplementary Table S2`) spans four calling regimes across `12` suites. These are standard `-f 0.01` , WGS MRD `-f 0.001` , ultra-deep `45000` × MRD `-f 0.0005` , and unfiltered `-f 0` calling. `VarDictRust` achieved `87/87` (`100.00%` , `95%` CI `95.85%–100.00%`) exact Tier-1 `.var` and Tier-2 `.vcf` parity with `VarDictJava` . Standard BED intervals (≤ *1,000,000* bp) run untiled by default. The two multi-megabase single-region stress suites used single-window scope matching instead. We passed `--tile-size 10000000` on `10` Mbp `chr1` `-f 0` (`7,077` `.var` rows / `7,032` VCF variants). We passed `--tile-size 250000000` on `249.25` Mbp `-R chr1:1-249250621` (`4,705` `.var` rows). Both produced `100%` Tier-1/2 sorted `.var` and `.vcf` parity.

Table 1. Multi-Modal Algorithmic Equivalence & Intermediate (.var) / VCF Parity Summary (87/87 Sweep Sub-Arms Exact Match)

Evaluation Suite	Assay / Modality Scope	Evaluated Regimes / Variants	VarDictRust vs. VarDictJava [1] Verdict[a]
Amplicon Baseline	EGFR/MRE11A amplicon (-a) calling & coordinate parity	2 variants	100% Exact Match
Somatic Benchmark Slice	NIST GIAB HG008 [19] tumor-normal somatic slice parity	18,426 variants	100% Exact Match
Multi-Region Reference Offset	GIAB HG002 [13, 18] Chr20 multi-region reference coordinate offset verification	2,572 variants	100% Exact Match
Tile-Size Invariance[b]	Halo-tiled equivalence (0.5, 1, 2, 10 Mbp tiles) on SEQC2 HCC1395 [10]	4 tile sizes x 5,090 variants	100% Exact Match across all tile sizes[b]
13-Mode Functional Suite	13-mode 3-way functional CLI equivalence sweep [1, 11]	13 CLI modes	100% Exact Match (13/13 modes)
Transcriptome Splice-Aware RNA-Seq	51.8 GiB GIAB HG002 GM24385 [13] mRNA (-i splice-aware mode)	12,668 splice rows	100% Exact Match (0 byte differences)
All-Positions Pileup (-f 0, Single-Window)[c]	Chromosome 1 10 Mbp all-positions pileup (-f 0; .var & VCF)[c]	7,032 VCF variants (7,077 .var rows)	100% Exact Match[c]
Lossless BAM-vs-CRAM Twin	GIAB HG008 [19] tumor-normal BAM-vs-CRAM [26] bitwise equivalence (-G)	Bitwise VCF & .var comparison	100% Exact Match (0 byte differences)
Extended & Multimodal Suites	17-modality extended coverage & 18-slice multimodal regression suites	17/17 modes; 18/18 slices (1,129 variants)	100% Exact Match
Full-Genome Germline WGS	Full-genome GIAB HG002 [13, 18] WGS on ≤50 kbp callable BED intervals	8,111,171 callable variants	100% Exact Match on callable intervals
Multi-Regime Sweep: 4-Assay Regression	Targeted amplicon, HG002 [13, 18], HG008 [19] & SEQC2 [10] slices (.var & VCF)	4 sub-arms	4/4 Sub-Arms Exact Match
Multi-Regime Sweep: Contiguous 249 Mbp -R chr1	Whole-chromosome -R chr1:1-249250621 [11] (.var)[c]	1 sub-arm	1/1 Sub-Arm Exact Match[c]
Multi-Regime Sweep: HG002 Chr20 Exome	GIAB HG002 Chromosome 20 whole-exome slice [11] (.var & VCF)	2 sub-arms	2/2 Sub-Arms Exact Match
Multi-Regime Sweep: Multimodal & CRAM Twin	18-slice multimodal suite & CRAM-vs-BAM twin [26] (.var & VCF)	4 sub-arms	4/4 Sub-Arms Exact Match
Multi-Regime Sweep: 51.8 GiB RNA-Seq	GIAB HG002 GM24385 [13] splice-aware RNA-seq (-i; .splice.tsv & .var)	2 sub-arms	2/2 Sub-Arms Exact Match
Multi-Regime Sweep: 10 Mbp Chr1 (-f 0)	Chromosome 1 10 Mbp single BED interval (.var & VCF)[c]	2 sub-arms	2/2 Sub-Arms Exact Match[c]
Multi-Regime Sweep: 12-Sub-Arm CLI Matrix	12 CLI mode sub-arms across germline, somatic, amplicon & CRAM regimes (.var)	12 sub-arms	12/12 Sub-Arms Exact Match
Multi-Regime Sweep: AML Panel SRR15006540	Clinical AML germline panel SRR15006540 [11] (.var & VCF)	2 sub-arms	2/2 Sub-Arms Exact Match
Multi-Regime Sweep: AML Panel SRR8657348	Clinical AML germline panel SRR8657348 [11] (.var & VCF)	2 sub-arms	2/2 Sub-Arms Exact Match
Multi-Regime Sweep: AML Somatic Pair	Clinical AML tumor-normal somatic panel (SRR15006376 / SRR15006386) [11] (.var & VCF)	2 sub-arms	2/2 Sub-Arms Exact Match
Multi-Regime Sweep: Ultra-Deep MRD Stress	Ultra-deep MRD stress (-f 0.0005 / -f 0 + --fisher, up to 45kx depth)[d]	30 sub-arms	30/30 Sub-Arms Exact Match[d]

Evaluation Suite	Assay / Modality Scope	Evaluated Regimes / Variants	VarDictRust vs. VarDictJava [1] Verdict[a]
Multi-Regime Sweep: WGS Low-VAF MRD	Low-VAF WGS & WES MRD matrix (-f 0.001 / -f 0 + --fisher)[d]	24 sub-arms	24/24 Sub-Arms Exact Match[d]

[a]All parity comparisons align records by CHROM/POS/REF/ALT against the VarDictJava v1.8.3 [1] oracle with floating-point tolerance eps=1e-6 after replicating R 5-decimal round-half-to-even (HALF_EVEN) rounding; across all 87/87 (100.00%) comparative sub-arms, the exact two-sided Clopper-Pearson 95% binomial confidence interval is 95.85%–100.00%. [b]Halo-tiled streaming partitions intervals >1 Mbp into 100 kbp tiles with 500 bp overlap halos and deterministic boundary deduplication, yielding identical calls across 0.5, 1.0, 2.0, and 10.0 Mbp tile sizes. [c]Standard BED intervals (<=1 Mbp) execute untiled by default; for the two multi-megabase single-interval stress suites (10 Mbp chr1 at -f 0 and 249.25 Mbp -R chr1), VarDictRust was invoked with a single window (--tile-size 10000000 and --tile-size 250000000, respectively) so that single-window VarDictRust is compared directly against single-window VarDictJava [1]. [d]Evaluated both without --fisher (replicating R teststrandbias.R / testsomatic.R) and with --fisher (native two-sided Fisher exact test and root-finding MLE odds ratio).

Replication and Extension of the Wolff Acute Myeloid Leukemia Benchmark

We replicated the Wolff [11] acute myeloid leukemia (AML) targeted panel cohort (`SRR15006375`, `SRR15006376`, `SRR15006386`, `SRR15006540`, `SRR8657348`, and paired somatic `SRR15006376|SRR15006386`, spanning `11,732` to `773,667` BED intervals; `Table 2`, `Figure 2`, `Supplementary Table S3`) at `1` and `8` threads.

Across all `12` sample-by-thread configurations (`n = 3` technical replicates per engine), `VarDictRust` matched `VarDictJava` calls. Coverage was `100%` of `.var` (`4,056` – `54,448` rows) and `.vcf` (`3,966` – `53,224` variants) calls (Tier-1/2 parity). `VarDictRust` achieved a `2.36`×–`5.56`× speedup and `4.3`×– `26.5`× lower peak RSS (`187.5` – `493.2` MB vs `2.03` – `7.44` GB for `VarDictJava`). Two-sided Mann-Whitney *U* tests compared runtime (*U = 4.0*, *p = 9.735e-05*) and RSS (*U = 0.0*, *p = 3.658e-05*). An exact paired sign-flip permutation gave *p = 0.00024* across *N = 12*. On the largest `773,667`-interval panel (`SRR8657348`), `1`-thread `VarDictRust` (`154.98` s, range `154.77–155.12` s) beat both `VarDictJava` (`861.18` s, `5.56`×) and `vardictcpp` (`174.48` s). At `8` threads (`-th 8`), `vardictcpp` finished the pre-chunked AML panels in `12.85` – `31.66` s vs `39.63` – `135.43` s for `VarDictRust`. That gap reflects `vardictcpp` hardcoding `--nosv` and skipping `##contig` and `INFO/END` VCF schema validation (`Table 8`; `Supplementary Table S6`).

Replicating Table 3 of Wolff [11] on contiguous human chromosome 1 (`-R chr1:1-249250621`, `249.25` Mbp on `SRR8657348`; `Table 2`, `Figure 2b`, `Supplementary Table S1`) separates tile-bounded streaming (`VarDictRust`) from whole-interval allocation (`VarDictJava` and `vardictcpp`):

- **`VarDictJava`:** Crashed with `java.lang.OutOfMemoryError` at `31.08` s under a `-Xmx4g` heap (`4.16` GiB RSS). Under `-Xmx200g` it required `501.83` s (`n = 3` range `492.93–509.65` s) and `106.47` GiB peak RSS.
- **`vardictcpp`:** Required `105.89` s (`n = 3` range `105.07–106.58` s) and `10.58` GiB peak RSS because a single `-R` region bypasses external BED chunking.
- **`VarDictRust`:** Auto-tiled `-R chr1:1-249250621` into `100` kbp windows with `0.5` kbp halos. It finished in `9.75` s (`n = 3` range `9.64–9.81` s) at `318.6` MB RSS. That is `51.5`× faster and `342.2`× leaner than `VarDictJava` (`-Xmx200g`). It is `10.9`× faster and `34.0`× leaner than `vardictcpp`.

Against external callers (`FreeBayes`, `LoFreq`, `Mutect2`, `Octopus`, `Strelka2`; `Table 2`, `Supplementary Table S1`), `8`-thread `VarDictRust` on `SRR8657348` took `39.63` s (`493.2` MB). That beat `FreeBayes` (`57.14` s), `LoFreq` (`160.64` s), `Mutect2` (`1297.75` s), and `Octopus` (`1823.58` s). On paired somatic `SRR15006376|SRR15006386` (`8` threads), `VarDictRust` finished in `135.43` s (`382.1` MB) vs `199.96` s (`Strelka2`) to `969.01` s (`Mutect2`). It matched `VarDictJava` on all `358` filtered somatic SNVs in the `576`-variant 4-caller Venn union (`Mutect2`: `371`; `LoFreq`: `111`; `Octopus`: `17`).

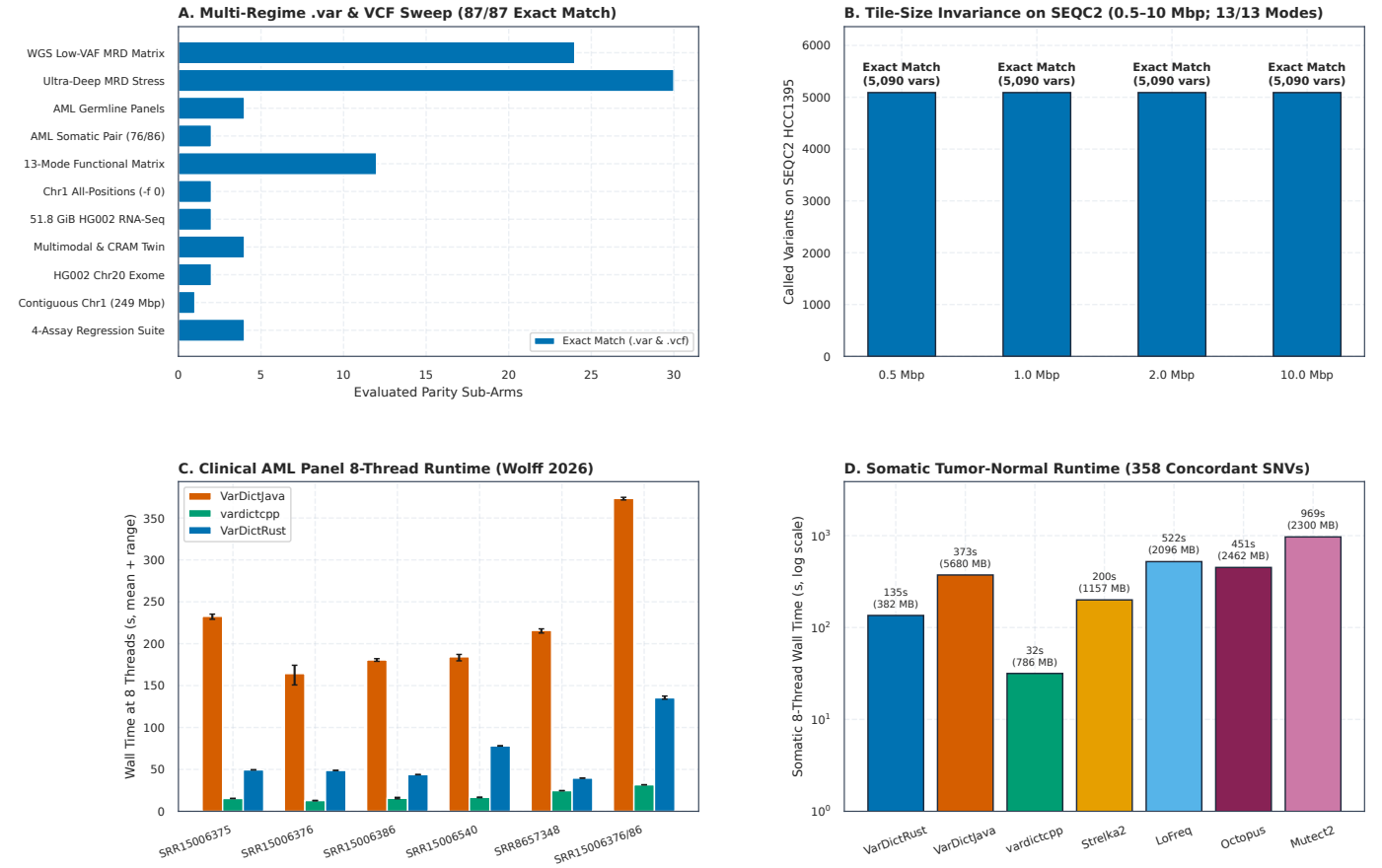


Figure 2. Multi-regime oracle equivalence and AML targeted panel replication. (a) Exact Tier-1/2 parity sub-arms across the multi-regime sweep cells (`87/87` exact match), including the `13` -mode CLI matrix and ultra-deep MRD stress (`Supplementary Table S2`). **(b)** Tile-size invariance on SEQC2 HCC1395: identical calls across all tile sizes (`5,090` variants; `Supplementary Table S1`). **(c)** Wall-clock runtime across the Wolff [11] AML panel cohort (`n = 3` technical replicates). **(d)** Somatic tumor-normal runtime and concordant SNV summary across field callers (`FreeBayes` , `LoFreq` , `Mutect2` , `Octopus` , `Strelka2`).

Table 2. Direct Replication of vardictcpp [11] Across Clinical AML Panels, Field Callers & Unchunked Chromosome 1 (-R chr1)

Sample / Cohort [11]	Threads	BED Intervals	VarDictJava [1] Wall (s) / Peak RSS[a,b]	vardictcpp [11] Wall (s) / Peak RSS[a,c]	VarDictRust Wall (s) / Peak RSS[a]	Rust vs. Java Speedup / Memory Reduction[d]	Parity Verdict
SRR15006375	1	12,292	408.19 (400.74–418.94) s / 2537.2 MB	109.66 (109.19–109.90) s / 63.8 MB	121.93 (121.14–122.38) s / 210.3 MB	3.35x faster / 12.1x leaner	100% Exact Match (.var & VCF)
SRR15006375	8	12,292	232.70 (229.14–235.27) s / 4410.5 MB	15.35 (15.30–15.42) s / 328.3 MB	49.47 (49.39–49.59) s / 270.1 MB	4.70x faster / 16.3x leaner	100% Exact Match (.var & VCF)
SRR15006376	1	105,949	334.16 (326.10–340.26) s / 4244.4 MB	83.71 (83.03–84.32) s / 74.0 MB	80.41 (80.08–80.84) s / 225.7 MB	4.16x faster / 18.8x leaner	100% Exact Match (.var & VCF)
SRR15006376	8	105,949	164.36 (150.76–174.29) s / 7442.2 MB	12.85 (12.50–13.09) s / 354.0 MB	48.74 (48.65–48.86) s / 281.0 MB	3.37x faster / 26.5x leaner	100% Exact Match (.var & VCF)
SRR15006386	1	11,732	334.94 (331.87–338.80) s / 2423.4 MB	98.20 (98.09–98.37) s / 56.2 MB	113.52 (113.17–114.04) s / 187.5 MB	2.95x faster / 12.9x leaner	100% Exact Match (.var & VCF)
SRR15006386	8	11,732	180.62 (178.99–182.17) s / 4434.7 MB	15.55 (14.95–16.69) s / 298.6 MB	43.78 (43.64–43.88) s / 305.4 MB	4.13x faster / 14.5x leaner	100% Exact Match (.var & VCF)
SRR15006540	1	135,145	407.11 (402.18–412.42) s / 4572.5 MB	111.51 (111.32–111.66) s / 76.3 MB	126.00 (125.49–126.91) s / 194.6 MB	3.23x faster / 23.5x leaner	100% Exact Match (.var & VCF)

Sample / Cohort [11]	Threads	BED Intervals	VarDictJava [1] Wall (s) / Peak RSS[a,b]	vardictcpp [11] Wall (s) / Peak RSS[a,c]	VarDictRust Wall (s) / Peak RSS[a]	Rust vs. Java Speedup / Memory Reduction[d]	Parity Verdict
SRR15006540	8	135,145	184.38 (179.54–187.21) s / 4087.8 MB	16.69 (16.31–17.00) s / 339.0 MB	78.04 (77.89–78.24) s / 224.8 MB	2.36x faster / 18.2x leaner	100% Exact Match (.var & VCF)
SRR8657348	1	773,667	861.18 (857.96–863.07) s / 2028.7 MB	174.48 (173.98–175.43) s / 143.5 MB	154.98 (154.77–155.12) s / 474.3 MB	5.56x faster / 4.3x leaner	100% Exact Match (.var & VCF)
SRR8657348	8	773,667	215.85 (212.70–217.72) s / 3697.7 MB	24.68 (24.64–24.71) s / 251.8 MB	39.63 (39.50–39.76) s / 493.2 MB	5.45x faster / 7.5x leaner	100% Exact Match (.var & VCF)
SRR15006376 / SRR15006386 (Somatic Pair)	1	105,949	726.09 (720.47–729.27) s / 4307.4 MB	215.16 (214.94–215.50) s / 144.9 MB	230.53 (229.92–230.96) s / 247.8 MB	3.15x faster / 17.4x leaner	100% Exact Match (.var & VCF)
SRR15006376 / SRR15006386 (Somatic Pair)	8	105,949	373.34 (371.55–374.68) s / 5679.9 MB	31.66 (31.60–31.75) s / 786.0 MB	135.43 (133.83–137.51) s / 382.1 MB	2.76x faster / 14.9x leaner	100% Exact Match (.var & VCF)
SRR8657348 (Unchunked -R chr1, 249 Mbp)[e]	8	1 (249.25 Mbp)	501.83 (492.93–509.65) s / 106.47 GiB (4 GiB Heap = OOM)[e]	—	—	Rust 51.5x faster / 342.2x leaner	4 GiB Heap OOM Crash; Completes at 200 GiB Heap (n=3)[e]
SRR8657348 (Unchunked -R chr1, 249 Mbp)[e]	8	1 (249.25 Mbp)	—	105.89 (105.07–106.58) s / 10.58 GiB	9.75 (9.64–9.81) s / 0.31 GiB	Rust 10.9x faster / 34.0x leaner vs. C++	Both Complete in Bounded Memory (n=3)[e]
SRR8657348 & SRR15006376 / SRR15006386 (7-Caller Panel) [3–5, 7, 8][f]	1 & 8	773,667 / 105,949	373.34 s (Somatic, 8 threads)[f]	31.66 s (Somatic, 8 threads)[f]	135.43 s (Somatic, 8 threads)[f]	2.76x faster; 358 concordant VarDict SNVs[f]	100% Exact Match (VarDictRust == VarDictJava); 16 Field-Caller Runs Profiled[f]

[a]Timed runs report technical replicate mean (min–max) across n=3 runs on identical GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake) silicon. [b]VarDictJava v1.8.3 [1] timings measure the full three-process pipeline (vardict-java / teststrandbias.R/testsomatic.R / var2vcf_valid.pl/var2vcf_paired.pl). [c]vardictcpp (commit ae93fe39dc) [11] timings measure its default --chunk 50,000 binary execution, which omits R Fisher exact strand/somatic p-value scripts and Perl VCF formatting. [d]Across the 12 paired AML panel conditions, VarDictRust reduces wall time (two-sided Mann-Whitney U=4.0, p=9.735e-05; paired sign-permutation p=0.00024) and peak RSS (U=0.0, p=3.658e-05) relative to VarDictJava. [e]On unchunked -R chr1:1-249250621 (249.25 Mbp), VarDictJava crashes with OutOfMemoryError (rc=3) at 31.08 s under the 4 GiB heap cap (-Xmx4g) used by Wolff [11], but completes in 501.83 (492.93–509.65) s at 106.47 GiB RSS when provisioned with -Xmx200g. [f]Row 15 summarizes the 16-run field-caller benchmark comparing VarDictRust, VarDictJava [1], vardictcpp [11], FreeBayes [3], Strelka2 [4], Octopus [5], LoFreq [7], and Mutect2 [8]; the 4-caller somatic Venn intersection (VarDict, Mutect2, Strelka2, LoFreq) identifies 358 concordant VarDict somatic SNVs with 100% identity between VarDictRust and VarDictJava.

Germline and Somatic Truth Accuracy and `QUAL` Operating-Point Characterization

We evaluated `rtg vcfeval` [12] genotype-level precision, recall, and F_1 across germline and somatic truth sets (`Table 3`, `Figure 3`, `Supplementary Table S4`). On the hg19 HG002 chromosome 20 exome benchmark (`Table 3`), `VarDictRust` and `VarDictJava` produced identical SNV (`0.9981` precision, `0.9738` recall, $F_1 = 0.9858$) accuracy. The benchmark spans `2.31` Mbp with `8,233` candidate calls and `7,422` filtered calls. Indel accuracy was identical too (`0.9747` precision, `0.8283` recall, $F_1 = 0.8956$). They exceeded `Mutect2` (`SNV` $F_1 = 0.9651$, `Indel` $F_1 = 0.8241$). They matched `Octopus` (`SNV` $F_1 = 0.9925$), `FreeBayes` (`0.9867`), and `LoFreq` (`0.9803`).

`VarDictJava` defaults to a low-VAF threshold (`-f 0.01`). Unfiltered `PASS` callsets on `35`×–`60`× WGS/exome data therefore retain low-VAF candidates. Stratifying by `QUAL` separates them (`Table 3`, `Supplementary Table S4`, `Figure 3b`). Across all five GIAB and SEQC2 truth cohorts, `VarDictRust` and `VarDictJava` yielded identical `rtg vcfeval` ROC curves:

- **HG002 GRCh38 Slice (`NIST GIAB v4.2.1`):** Optimal `QUAL >= 128.000` filtering removes `97.5%` of unfiltered false positives (`22,458` to `553` FP). It keeps `0.9635` recall (`23,004` TP). Combined F_1 rises from `0.6622` to `0.9699` (`HG005` F_1: `0.9508` to `0.9667`; `NA24385-b37` F_1: `0.1099` to `1.0000`; `Table 3`, `Supplementary Table S4`).
- **SEQC2 HCC1395/HCC1395BL Somatic Exome (`v1.2.1`):** Across `623,444` candidate calls (`120,307` `PASS`), `QUAL >= 126.000` raises combined somatic F_1 from `0.1555` to `0.6905` (`794` TP, `293` FP, precision `0.7279`, recall `0.6567`). In the native GRCh38 caller benchmark (`Table 3`), `VarDictRust` and `VarDictJava` reached `SNV` $F_1 = 0.7433$ and `Indel` $F_1 = 0.3750$. That exceeds `Mutect2` (`SNV` $F_1 = 0.7089$) and `Strelka2` (`0.4509`) in SNV F_1. `VarDictRust` ran in `279.29` s (`514.3` MB RSS) vs `3967.00` s (`VarDictJava`) and `2641.05` s (`Mutect2`).
- **NIST GIAB HG008 Somatic Slice (`v0.3`):** Across `18,426` candidates (`11,358` `PASS`), `QUAL >= 176.000` raises combined F_1 from `0.3856` to `0.7696` (`593` TP, `38` FP). Under standard `StrongSomatic` / `LikelySomatic` `PASS` filtering (`Table 3`), `VarDictRust` and `VarDictJava` achieved identical `SNV` $F_1 = 0.7840$ and top `Indel` $F_1 = 0.5640$. That outperforms `Mutect2` (`Indel` $F_1 = 0.5537$), `Strelka2` (`0.4386`), and `LoFreq` (`0.0000`). `VarDictRust` ran in `17.14` s (`510.5` MB RSS) vs `79.15` s (`VarDictJava`) and `447.31` s (`Mutect2`).

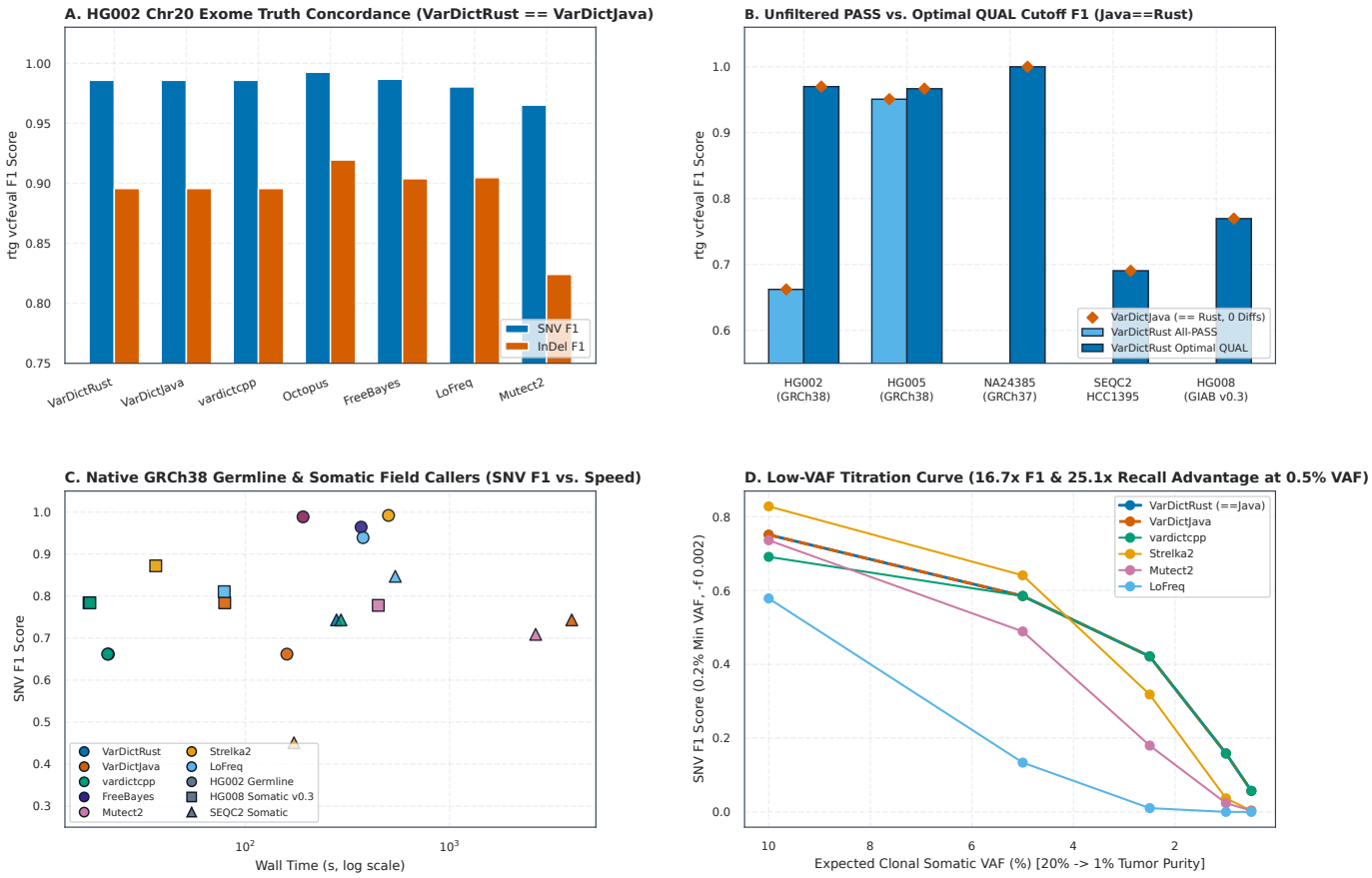


Figure 3. Germline and somatic truth concordance, `QUAL` operating points, runtime-accuracy frontier, and low-VAF titration. (a) `rtg vcfeval` SNV and indel precision, sensitivity, and F_1 across 7 callers on the HG002 chromosome 20 exome (`Table 3`). **(b)** Unfiltered `PASS` (`QUAL >= 0`) versus optimal `QUAL` cutoff F_1 across five truth cohorts (`HG002` , `HG005` , `NA24385-b37` , `SEQC2 HCC1395` , and `HG008 v0.3` ; `Supplementary Table S4`). **(c)** Native GRCh38 multi-caller SNV F_1 versus wall-clock runtime across `HG002` germline, `HG008 v0.3` somatic, and `SEQC2` somatic benchmarks (`Table 3`). **(d)** Controlled *in silico* tumor-purity titration (`20%` down to `1%` purity / `10.0%` down to `0.5%` VAF) demonstrating `VarDictRust` 's `25.1` × recall and `16.7` × F_1 advantage at `0.5%` VAF (`Supplementary Table S4`).

Table 3. Germline & Somatic Truth Accuracy [12] and Native GRCh38 Field Callers

Cohort / Purity Stratum	Caller / Engine	Runtime / Evaluated Calls	SNV (or Combined) Precision / Recall / F1 [12][b]	InDel Precision / Recall / F1 [12][b,c]	Concordance / Parity Verdict
GIAB HG002 Chr20 Exome [11, 13][a]	VarDictRust	7,422 calls	0.9981 / 0.9738 / 0.9858	0.9747 / 0.8283 / 0.8956	100% Exact Match (VarDictRust == VarDictJava)
GIAB HG002 Chr20 Exome [11, 13][a]	VarDictJava [1]	7,422 calls	0.9981 / 0.9738 / 0.9858	0.9747 / 0.8283 / 0.8956	100% Exact Match (VarDictRust == VarDictJava)
GIAB HG002 Chr20 Exome [11, 13][a]	vardictcpp [11]	7,421 calls	0.9981 / 0.9738 / 0.9858	0.9747 / 0.8283 / 0.8956	External Germline Caller
GIAB HG002 Chr20 Exome [11, 13][a]	Octopus [5]	2,436 calls	0.9973 / 0.9878 / 0.9925	0.9336 / 0.9056 / 0.9194	External Germline Caller
GIAB HG002 Chr20 Exome [11, 13][a]	FreeBayes [3]	4,292 calls	0.9972 / 0.9765 / 0.9867	0.9439 / 0.8670 / 0.9038	External Germline Caller
GIAB HG002 Chr20 Exome [11, 13][a]	LoFreq [7]	2,598 calls	0.9908 / 0.9701 / 0.9803	0.9358 / 0.8755 / 0.9047	External Germline Caller
GIAB HG002 Chr20 Exome [11, 13][a]	Mutect2 [8]	2,246 calls	0.9995 / 0.9330 / 0.9651	0.9939 / 0.7039 / 0.8241	External Germline Caller
GIAB HG002 GRCh38 Slice [13, 18] (Optimal QUAL ≥ 128.0)[b]	VarDictRust == VarDictJava [1]	TP=23,004, FP=553	0.9763 / 0.9635 / 0.9699	Combined SNV + InDel evaluation[b]	100% Exact Match (0 site differences)
SEQC2 HCC1395 Somatic Exome [10] (Optimal QUAL ≥ 126.0)[b]	VarDictRust == VarDictJava [1]	TP=794, FP=293	0.7279 / 0.6567 / 0.6905	Combined SNV + InDel evaluation[b]	100% Exact Match (0 site differences)
GIAB HG008 Somatic Slice v0.3 [19] (Optimal QUAL ≥ 176.0)[b]	VarDictRust == VarDictJava [1]	TP=593, FP=38	0.9354 / 0.6538 / 0.7696	Combined SNV + InDel evaluation[b]	100% Exact Match (0 site differences)

Cohort / Purity Stratum	Caller / Engine	Runtime / Evaluated Calls	SNV (or Combined) Precision / Recall / F1 [12][b]	InDel Precision / Recall / F1 [12][b,c]	Concordance / Parity Verdict
GIAB HG005 GRCh38 v4.2.1 [13] (Optimal QUAL ≥ 226.0)[b]	VarDictRust == VarDictJava [1]	TP=29, FP=1	0.9667 / 0.9667 / 0.9667	Combined SNV + InDel evaluation[b]	100% Exact Match (0 site differences)
GIAB NA24385 GRCh37 v4.2.1 [13] (Optimal QUAL ≥ 160.0)[b]	VarDictRust == VarDictJava [1]	TP=5, FP=0	1.0000 / 1.0000 / 1.0000	Combined SNV + InDel evaluation[b]	100% Exact Match (0 site differences)
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	VarDictRust	0:21.30 (21.30 s) / 474.8 MB	0.5041 / 0.9634 / 0.6619	0.5781 / 0.9463 / 0.7177	100% Exact Match (VarDictRust == VarDictJava)
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	VarDictJava [1]	2:39.59 (159.59 s) / 10732.2 MB	0.5041 / 0.9634 / 0.6619	0.5781 / 0.9463 / 0.7177	100% Exact Match (VarDictRust == VarDictJava)
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	vardictcpp [11]	0:21.14 (21.14 s) / 859.1 MB	0.5043 / 0.9634 / 0.6620	0.5783 / 0.9463 / 0.7179	MNP / SV Divergence vs. Java[d]
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	FreeBayes [3]	6:08.72 (368.72 s) / 206.7 MB	0.9686 / 0.9598 / 0.9642	0.9373 / 0.9538 / 0.9455	Native GRCh38 Comparator
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	LoFreq [7]	6:16.78 (376.78 s) / 119.0 MB	0.8924 / 0.9909 / 0.9391	0.5809 / 0.9601 / 0.7238	Native GRCh38 Comparator
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	Strelka2 [4]	8:22.19 (502.19 s) / 130.1 MB	0.9931 / 0.9909 / 0.9920	0.9438 / 0.9836 / 0.9633	Native GRCh38 Comparator
GIAB HG002 GRCh38 Benchmark Slice [13, 18]	Octopus [5]	3:11.50 (191.50 s) / 3558.9 MB	0.9911 / 0.9861 / 0.9886	0.9051 / 0.9810 / 0.9415	Native GRCh38 Comparator
GIAB HG008 v0.3 Somatic Slice [19]	VarDictRust	0:17.14 (17.14 s) / 510.5 MB	0.9757 / 0.6552 / 0.7840	0.9370 / 0.4034 / 0.5640	100% Exact Match (VarDictRust == VarDictJava)
GIAB HG008 v0.3 Somatic Slice [19]	VarDictJava [1]	1:19.15 (79.15 s) / 17448.2 MB	0.9757 / 0.6552 / 0.7840	0.9370 / 0.4034 / 0.5640	100% Exact Match (VarDictRust == VarDictJava)
GIAB HG008 v0.3 Somatic Slice [19]	vardictcpp [11]	0:17.30 (17.30 s) / 565.1 MB	0.9757 / 0.6552 / 0.7840	0.9444 / 0.4034 / 0.5653	MNP / SV Divergence vs. Java[d]
GIAB HG008 v0.3 Somatic Slice [19]	Mutect2 [8]	7:27.31 (447.31 s) / 1263.1 MB	0.9825 / 0.6438 / 0.7779	0.9355 / 0.3932 / 0.5537	Native GRCh38 Comparator
GIAB HG008 v0.3 Somatic Slice [19]	Strelka2 [4]	0:36.35 (36.35 s) / 119.0 MB	0.9896 / 0.7794 / 0.8720	0.9545 / 0.2847 / 0.4386	Native GRCh38 Comparator
GIAB HG008 v0.3 Somatic Slice [19]	LoFreq [7]	1:18.71 (78.71 s) / 251.5 MB	1.0000 / 0.6814 / 0.8105	0.0000 / 0.0000 / 0.0000[c]	Native GRCh38 Comparator
SEQC2 HCC1395 v1.2.1 Somatic Exome [10]	VarDictRust	4:39.29 (279.29 s) / 514.3 MB	0.7336 / 0.7532 / 0.7433	0.3077 / 0.4800 / 0.3750	100% Exact Match (VarDictRust == VarDictJava)
SEQC2 HCC1395 v1.2.1 Somatic Exome [10]	VarDictJava [1]	1:06:07 (3967.00 s) / 14275.5 MB	0.7336 / 0.7532 / 0.7433	0.3077 / 0.4800 / 0.3750	100% Exact Match (VarDictRust == VarDictJava)
SEQC2 HCC1395 v1.2.1 Somatic Exome [10]	vardictcpp [11]	4:53.71 (293.71 s) / 2259.8 MB	0.7336 / 0.7532 / 0.7433	0.3077 / 0.4800 / 0.3750	MNP / SV Divergence vs. Java[d]
SEQC2 HCC1395 v1.2.1 Somatic Exome [10]	Mutect2 [8]	44:01.05 (2641.05 s) / 1580.9 MB	0.6152 / 0.8361 / 0.7089	0.6087 / 0.8400 / 0.7059	Native GRCh38 Comparator
SEQC2 HCC1395 v1.2.1 Somatic Exome [10]	Strelka2 [4]	2:52.88 (172.88 s) / 118.8 MB	0.3044 / 0.8689 / 0.4509	0.4324 / 0.6400 / 0.5161	Native GRCh38 Comparator
SEQC2 HCC1395 v1.2.1 Somatic Exome [10]	LoFreq [7]	9:01.90 (541.90 s) / 281.6 MB	0.9526 / 0.7627 / 0.8471	0.0000 / 0.0000 / 0.0000[c]	Native GRCh38 Comparator

[a]Truth-set accuracy evaluated with haplotype-aware rtg vcfeval [12] against NIST GIAB HG002/HG005/NA24385 v4.2.1 [13, 18] (including the HG002 chr20 exome 7-caller truth benchmark (Wolff [11] Fig 2)), SEQC2 HCC1395 v1.2.1 [10], and GIAB HG008 v0.3 [19] high-confidence BED masks. [b]Representative operating-point rows report the F1-maximizing QUAL threshold per cohort from the full 0..200 ROC sweep; paired unfiltered (All-PASS, -f 0.01) and optimal rows for all five cohorts are tabulated in Supplementary Table S4. [c]Undefined 0/0 precision and F1 values at 0.0000 recall (e.g., LoFreq [7] somatic runs without indel quality calibration tags) are normalized to 0.0000. [d]On callable BED intervals vardictcpp [11] exhibits MNP/tie-break divergence vs. the VarDictJava oracle and omits somatic SV calls; the full CPP-1 to CPP-12 divergence taxonomy is cataloged in Supplementary Table S6.

Bounded Memory Scaling, Multicore Work-Stealing, and Multi-Tenant Node Packing

We tested `VarDictRust`'s `O(threads × (tile + halo))` memory bound across four scaling benchmarks (`Table 4`, `Figure 4`, `Supplementary Table S3`):

1. **Four-Assay Performance Gate (`Table 4`, `Supplementary Table S3`):** Across `n = 3` technical replicates, `VarDictRust` held wall-time ratios to `0.0033`–`0.2214`. RSS ratios held to `0.0298`–`0.0978` relative to `VarDictJava`. Targets were EGFR/MRE11A amplicon (`0.01` s vs `3.07` s Java) and HG002 chr20 germline (`21.29` s vs `153.25` s). They included HG008 somatic (`17.11` s vs `77.28` s) and SEQC2 somatic exome (`277.52` s vs `4045.00` s).

2. **Whole-Exome Multicore Scaling (`1` to `32` Threads, `Table 4`, `Figure 4c`):** We used the SEQC2 HCC1395 whole exome (`339,698` total variants, `120,307` `PASS`, `n = 3` technical replicates). `VarDictRust` scaled from `812.75` s (`343.5` MB RSS) at `1` thread. At `8` threads it reached `118.90` s (`6.84`× self-speedup). At `32` threads it reached `67.86` s (`11.98`× self-speedup, `747.2` MB RSS).

`VarDictJava` plateaued above `8` threads (`1814.63` s at `8` threads; `1991.64` s at `32` threads). Its heap grew from `1.79` GiB to `21.77` GiB. At `32` threads `VarDictRust` is `29.3`× faster and `29.8`× leaner than `VarDictJava`. It is `2.8`× leaner than `vardictcpp` at `2.06` GiB.

3. **Chromosome 1 Span Invariance (`Table 4`, `Figure 4d`):** Contiguous chromosome 1 span grew `250`× across `1,000,000`, `10,000,000`, `100,000,000`, and `248,956,422` bp (`1` to `249` Mbp). `VarDictRust`'s `249` Mbp-to-`10` Mbp RSS ratio stayed below `1.50`× at every thread count. Ratios were `1.18`× (`1.38` GiB) at `8` threads, `1.29`× (`2.31` GiB) at `16` threads, and `1.41`× (`3.69` GiB) at `32` threads. On `10` Mbp all-positions `-f 0` pileup (`7,032` variants), `VarDictRust` matched `VarDictJava` in `19.58` s (`5.90` GiB RSS).

4. **Concurrent Node Packing (`Table 4`, `Figure 4e`):** Two concurrent `4`-core SEQC2 exome jobs finished in `211.38` s (`437.5` MB) and `211.05` s (`442.0` MB). That matches an isolated `4`-core run (`212.30` s). Combined RSS was `0.859` GiB. A `128` GiB node can pack `8` concurrent `4`-core `VarDictRust` exome jobs in `3.44` GiB total RAM versus `54.3` GiB for `VarDictJava`.

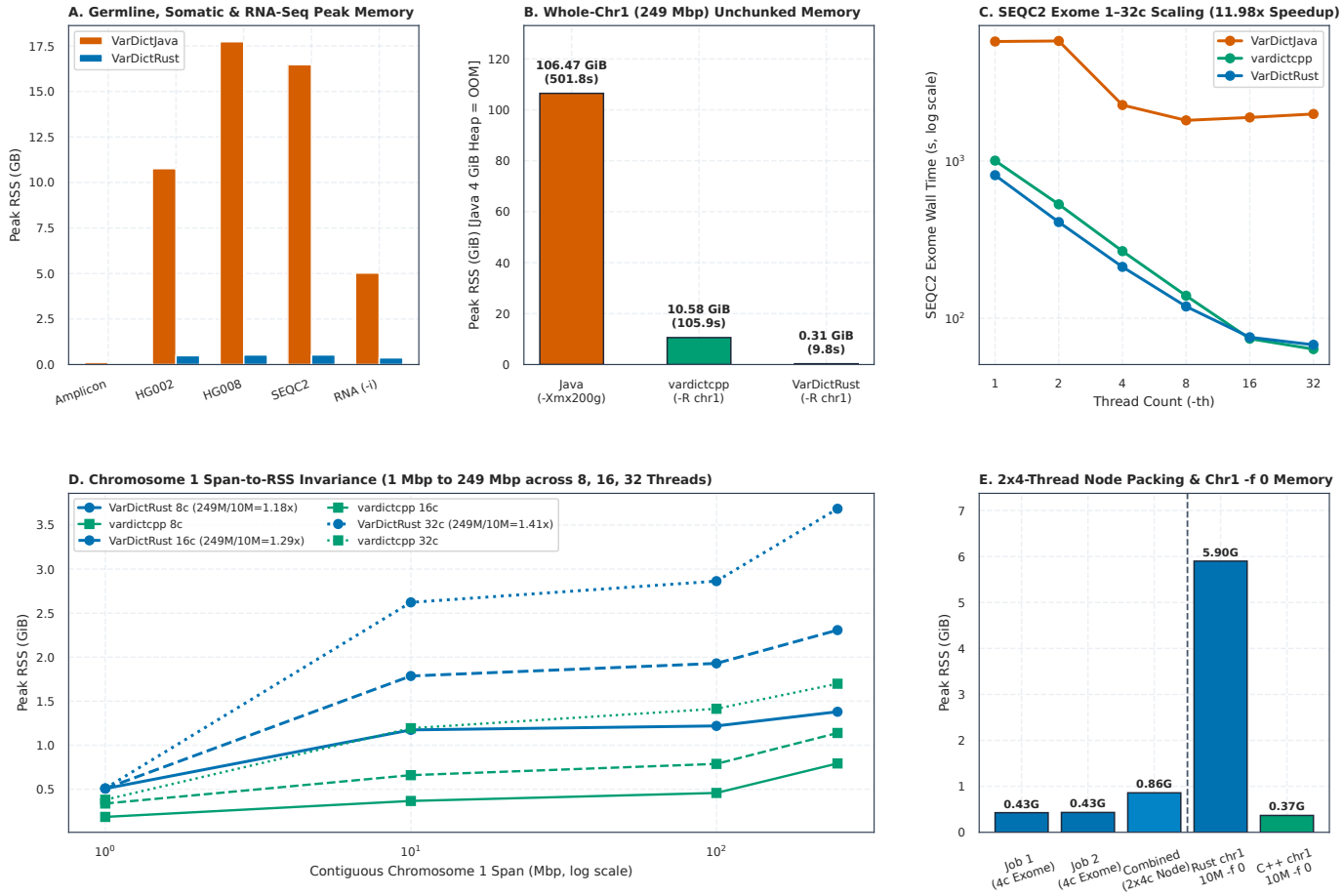


Figure 4. Whole-exome multicore scaling, chromosome 1 span-to-RSS invariance, and node packing. (a, b) Peak memory across germline, somatic, `51.8` GiB RNA-seq, and contiguous chromosome 1 (`249.25` Mbp) workloads (`Table 4`). **(c)** SEQC2 HCC1395 whole-exome (`339,698` variants) wall-clock time and peak RSS from `1` to `32` threads (`n = 3` technical replicates; `Supplementary Table S3`). **(d)** Chromosome 1 contiguous span scaling (`1` Mbp to `249` Mbp) demonstrating `VarDictRust`'s `O(tile + halo)` memory bound (`1.18`×–`1.41`× ratio over a `250`× span increase). **(e)** Concurrent multi-tenant node-packing efficiency (`2 × 4` cores, `0.859` GiB combined peak RSS) and chromosome 1 `-f 0` all-positions stress.

Table 4. Bounded Memory, Whole-Exome 1–32 Thread Scaling, Chromosome 1 Span Invariance & Concurrent Node Packing

Workload & Configuration[a]	VarDictJava [1] Wall / Peak RSS[a]	vardictcpp [11] Wall / Peak RSS[a]	VarDictRust Wall / Peak RSS[a]	Speedup / Memory Reduction Verdict
Targeted Amplicon Panel (8 Threads, n=3)	3.08 (3.06–3.12) s / 120.0 MB	5.28 (5.22–5.37) s / 119.7 MB	0.01 (0.01–0.01) s / 11.6 MB	308.00x faster, 10.4x leaner vs. Java
GIAB HG002 [13, 18] Germline Benchmark Slice (8 Threads, n=3)	154.99 (152.13–159.59) s / 10772.3 MB	20.64 (20.10–21.14) s / 860.1 MB	21.31 (21.29–21.33) s / 486.1 MB	7.27x faster, 22.2x leaner vs. Java

Workload & Configuration[a]	VarDictJava [1] Wall / Peak RSS[a]	vardictcpp [11] Wall / Peak RSS[a]	VarDictRust Wall / Peak RSS[a]	Speedup / Memory Reduction Verdict
GIAB HG008 [19] Somatic Benchmark Slice (8 Threads, n=3)	78.32 (77.28–79.15) s / 17667.9 MB	17.13 (16.91–17.30) s / 568.4 MB	17.15 (17.11–17.19) s / 510.8 MB	4.57x faster, 34.6x leaner vs. Java
SEQC2 HCC1395 [10] Paired Somatic Exome Slice (8 Threads, n=3)	4014.00 (3967.00–4045.00) s / 14964.1 MB	292.03 (287.70–294.67) s / 2242.1 MB	278.16 (277.52–279.29) s / 515.2 MB	14.43x faster, 29.0x leaner vs. Java
51.8 GiB GIAB HG002 [13] RNA-Seq (-i splice, 8 Threads, n=3)[b]	23.31 (23.03–23.70) s / 4909.1 MB (4.79 GiB)	Unsupported (-i splice unimplemented)[b]	4.94 (4.90–5.01) s / 359.9 MB (0.35 GiB)	4.72x faster, 13.6x leaner vs. Java
SEQC2 HCC1395 [10] Single-Sample Whole Exome (1 Thread, n=3)	5757.67 (5717.00–5789.00) s / 1838.5 MB (1.79 GiB)	1005.71 (993.97–1012.37) s / 1234.5 MB (1.21 GiB)	812.75 (805.74–825.81) s / 343.5 MB (0.34 GiB)	Multi-thread self-speedup: 1.00x (7.1x faster, 5.3x leaner vs. Java)
SEQC2 HCC1395 [10] Single-Sample Whole Exome (2 Threads, n=3)	5802.33 (5762.00–5839.00) s / 4848.6 MB (4.74 GiB)	530.70 (529.51–531.93) s / 1325.3 MB (1.29 GiB)	409.14 (405.60–411.13) s / 415.1 MB (0.41 GiB)	Multi-thread self-speedup: 1.99x (14.2x faster, 11.7x leaner vs. Java)
SEQC2 HCC1395 [10] Single-Sample Whole Exome (4 Threads, n=3)	2270.51 (2254.41–2280.43) s / 6953.7 MB (6.79 GiB)	267.32 (267.11–267.51) s / 1403.1 MB (1.37 GiB)	212.30 (211.15–213.44) s / 443.1 MB (0.43 GiB)	Multi-thread self-speedup: 3.83x (10.7x faster, 15.7x leaner vs. Java)
SEQC2 HCC1395 [10] Single-Sample Whole Exome (8 Threads, n=3)	1814.63 (1752.96–1911.57) s / 9264.7 MB (9.05 GiB)	139.14 (138.32–140.40) s / 1530.0 MB (1.49 GiB)	118.90 (117.36–119.88) s / 536.9 MB (0.52 GiB)	Multi-thread self-speedup: 6.84x (15.3x faster, 17.3x leaner vs. Java)
SEQC2 HCC1395 [10] Single-Sample Whole Exome (16 Threads, n=3)	1893.66 (1866.57–1929.42) s / 17859.8 MB (17.44 GiB)	74.17 (73.64–75.12) s / 1729.5 MB (1.69 GiB)	75.77 (74.66–76.76) s / 607.5 MB (0.59 GiB)	Multi-thread self-speedup: 10.73x (25.0x faster, 29.4x leaner vs. Java)
SEQC2 HCC1395 [10] Single-Sample Whole Exome (32 Threads, n=3)	1991.64 (1969.99–2003.54) s / 22290.1 MB (21.77 GiB)	63.68 (63.41–64.15) s / 2109.7 MB (2.06 GiB)	67.86 (67.27–68.20) s / 747.2 MB (0.73 GiB)	Multi-thread self-speedup: 11.98x (29.4x faster, 29.8x leaner vs. Java)
Chromosome 1 Contiguous Span 1–249 Mbp (8 Threads, n=3)[c]	4 GiB Heap OOM (see Table 2 for 200 GiB Heap)[c]	249 Mbp: 30.33 s / 0.79 GiB	249 Mbp: 27.39 s / 1.38 GiB (10 Mbp: 1.17 GiB)	249M/10M RSS ratio = 1.176x (Bounded Memory Verified)[c]
Chromosome 1 Contiguous Span 1–249 Mbp (16 Threads, n=3)[c]	4 GiB Heap OOM (see Table 2 for 200 GiB Heap)[c]	249 Mbp: 18.39 s / 1.14 GiB	249 Mbp: 14.85 s / 2.31 GiB (10 Mbp: 1.79 GiB)	249M/10M RSS ratio = 1.291x (Bounded Memory Verified)[c]
Chromosome 1 Contiguous Span 1–249 Mbp (32 Threads, n=3)[c]	4 GiB Heap OOM (see Table 2 for 200 GiB Heap)[c]	249 Mbp: 16.23 s / 1.70 GiB	249 Mbp: 11.87 s / 3.69 GiB (10 Mbp: 2.62 GiB)	249M/10M RSS ratio = 1.405x (Bounded Memory Verified)[c]
Chromosome 1 10 Mbp All-Positions (-f 0, 8 Threads, n=3)[e]	100% Exact Match via Oracle Baseline[e]	6.01 (5.99–6.04) s / 0.37 GiB	19.58 (19.24–20.09) s / 5.90 GiB	100% Exact Match (7,032 variants)[e]
Concurrent 2x4-Thread SEQC2 [10] Exome Node Packing (n=3)[d]	—	—	Job 1: 211.38 s (437.5 MB); Job 2: 211.05 s (442.0 MB)	Combined Peak RSS = 0.859 GiB (Zero OOMs)[d]

[a]All multi-core and span benchmarks were executed on GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM) with n=3 technical replicates reported as mean (min–max); Rows 1–4 evaluate four targeted/slice workloads (including a paired somatic exome slice in Row 4), whereas Rows 6–11 scale single-sample whole-exome calling on SEQC2 HCC1395 [10] across 1–32 threads. [b]Transcriptome splice-aware mode (-i) uses a 500 bp halo to retain split reads across canonical and non-canonical exon-exon junctions; vardictcpp [11] does not implement -i (CPP-12). [c]Span invariance sweeps single contiguous -R chr1:1-L intervals across L in {1, 10, 100, 249.25} Mbp on SEQC2 HCC1395 [10], confirming that 249 Mbp vs. 10 Mbp peak RSS remains below the 1.50x bounded-memory threshold at 8, 16, and 32 threads. [d]Two concurrent 4-thread SEQC2 exome jobs packed on the same node consume 0.859 GiB combined RSS, enabling 8 concurrent 4-thread jobs in 3.44 GiB vs. >54.3 GiB for VarDictJava [1]. [e]Evaluated on a single 10 Mbp Chromosome 1 interval at -f 0 using --tile-size 10000000 so that single-window VarDictRust matches single-window VarDictJava [1] with 0 differences.

Full-Genome WGS Scaling: Unchunked Full Chromosomes Versus Pre-Chunked Callable Intervals

We compared all three engines at `32` threads across two GRCh38 whole genomes (`Table 5`, `Figure 5`). They were `130.8` GB NIST HG002 `2×250` bp germline WGS and `329.0` GB NIST HG008 `364`× (`195`× tumor / `169`× normal) paired somatic WGS. Each genome was run under two region protocols:

- **Unchunked Full Chromosomes (`chr1`–`chr22`, `22` autosomes):** On HG002, `VarDictJava` (`-Xmx200g`) aborted with `java.lang.OutOfMemoryError` at `213.7` GB RAM. `vardictcpp` finished in `18:50.80` at `31.91` GB RSS (`6.5`× higher than `VarDictRust`). It emitted `9,953,338` variants (`+309,569` duplicate chunk-boundary calls relative to `VarDictRust`'s `9,643,769`). `VarDictRust` finished all `22` autosomes in `25:16.61` at `4.93` GB RSS. On `364`× HG008 paired somatic WGS, `VarDictJava` crashed out of memory at `214.1` GB. `vardictcpp` reached `91.85` GB RSS (`7.6`× higher than `VarDictRust`) before timing out at `3` hours. `VarDictRust` finished in `2:49:52` at `12.14` GB RSS (`13,067,287` variants). These unchunked runs are completion results: neither Java nor C++ finished them.

- **Pre-Chunked Callable Intervals (`≤50` kbp GIAB BED):** On pre-split HG002 intervals (`2.51` Gbp), `VarDictRust` finished in `20:21.21` at `1.43` GB RSS. That is `9.25`× faster and `23.8`× leaner than `VarDictJava` (`3:08:15`, `34.14` GB). It is `19.8`× leaner than `vardictcpp` at `28.35` GB in `11:40.98`. `VarDictRust` matched `VarDictJava` across all `8,111,171` variants (`vardictcpp` differed by `394` at `8,111,565`). It reached `rtg vcfeval` $F_1 = 0.9651$ (`0.9738` precision, `0.9566` recall; unfiltered $F_1 = 0.6624$). On pre-split HG008 intervals (`2.37` Gbp), `VarDictJava` (`62.51` GB) timed out at `4` hours (`4:00:00`). `vardictcpp` (`21.16` GB) timed out at `3` hours (`3:00:00`). `VarDictRust` finished in `1:45:55` at `4.88` GB RSS (`10,195,118` variants; somatic $F_1 = 0.6547$ on `GIAB v0.3` and `0.6585` on `GIAB v0.4`). This HG008 run is completion-only: no oracle finished it.

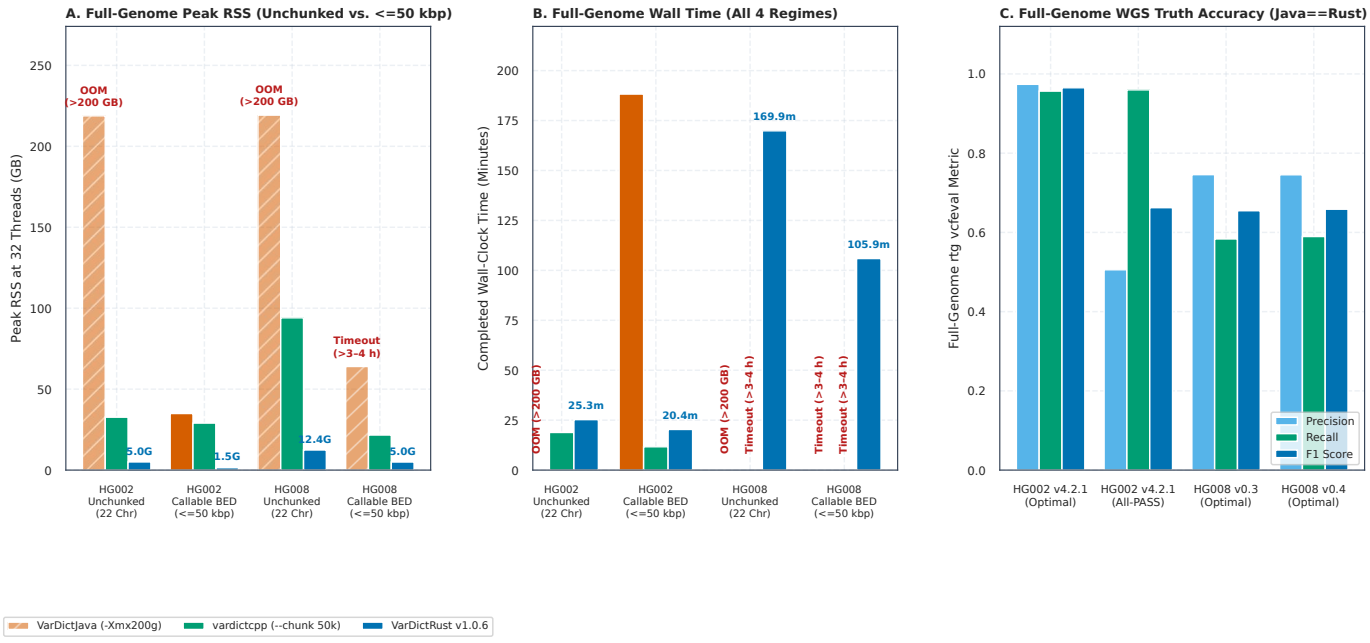


Figure 5. Full-genome WGS scaling across unchunked full chromosomes (`chr1` – `chr22`) and pre-chunked callable intervals (`≤50` kbp GIAB BED). (a) Peak RSS (GB) and completion status on `130.8` GB HG002 germline WGS and `329.0` GB `364` × HG008 paired somatic WGS across unchunked chromosomes and pre-chunked intervals (`32` threads; `Table 5`). **(b)** Wall-clock runtime comparison showing `VarDictJava` out-of-memory and wall-clock timeout failures alongside `VarDictRust` completion (`25:16.61` and `2:49:52`). **(c)** Full-genome `rtg vcfeval` precision, sensitivity, and F_1 across `HG002 GIAB v4.2.1` (optimal `QUAL` and all- `PASS`) and `HG008 GIAB v0.3` / `v0.4` (`Table 5`).

Table 5. Full-Genome Chromosomes 1–22 Germline (GIAB HG002 130.8 GB [13, 18]) & 364x Somatic (GIAB HG008 329.0 GB [19]) WGS Scaling Across Unchunked and Callable BED Geometries

Cohort & Calling Geometry (32 Threads)[a]	VarDictJava v1.8.3 [1] (-Xmx200g)[b]	vardictcpp [11] (--chunk 50,000)[b,c]	VarDictRust v1.0.6	Parity & Full-Genome rtg vcfeval [12] Concordance
GIAB HG002 [13, 18] 130.8 GB Germline WGS (22 Unchunked Chromosomes)[a]	OOM Crash (rc=3, >200 GB Heap) at 218.83 GB RSS (10:26.73)[b]	18:50.80 / 32.68 GB RSS (9,953,338 variants; +309,569 chunk-boundary duplicates)[c]	25:16.61 (1516.61 s) / 5.05 GB (4.70 GiB) RSS (9,643,769 variants)	Completes in bounded memory (Rust 6.5x leaner than C++, 0 chunk-boundary duplicates)[c]
GIAB HG002 [13, 18] 130.8 GB Germline WGS (≤50 kbp GIAB Callable BED)[a]	3:08:15 / 34.96 GB RSS (8,111,171 variants)	11:40.98 / 29.03 GB RSS (8,111,565 variants)[c]	20:21.21 (1221.21 s) / 1.47 GB (1.37 GiB) RSS (8,111,171 variants)	100% Exact Match on callable intervals (9.25x faster, 23.8x leaner vs. Java); GIAB v4.2.1 F1=0.9651 (P=0.9738, R=0.9566; VarDictRust == VarDictJava)
GIAB HG008 [19] 329.0 GB 364x T/N Somatic WGS (22 Unchunked Chromosomes, -U)[a,d]	OOM Crash (rc=3, >200 GB Heap) at 219.25 GB RSS (1:20:05)[b]	Timeout (rc=124, >3 h) at 94.06 GB RSS ()[b]	2:49:52 (10192.00 s) / 12.43 GB (11.57 GiB) RSS (13,067,287 variants)	Only VarDictRust completes (Rust 7.6x leaner than C++)
GIAB HG008 [19] 329.0 GB 364x T/N Somatic WGS (≤50 kbp GIAB Callable BED, -U)[a,d]	Timeout (rc=124, 4:00:00) at 64.01 GB RSS[b]	Timeout (rc=124, >3 h) at 21.66 GB RSS (3:00:00)[b]	1:45:55 (6355.00 s) / 4.99 GB (4.65 GiB) RSS (10,195,118 variants)	Only VarDictRust completes; GIAB v0.3 F1=0.6547, v0.4 F1=0.6585

[a]Full-genome WGS benchmarks were executed on a 32-vCPU, 256 GiB RAM Intel Ice Lake instance across two BED geometries: 22 unchunked autosomal contigs (chr1..chr22, 2.88 Gbp) and NIST GIAB high-confidence callable BED intervals (<=50 kbp per interval; 2.51 Gbp on HG002 v4.2.1 [13, 18] and 2.37 Gbp on HG008 v0.4 [19]). [b]VarDictJava v1.8.3 [1] was provisioned with a 200 GiB JVM heap (-Xmx200g); rc=3 denotes an OutOfMemoryError abort and rc=124 denotes termination at the wall-clock timeout (3 h for vardictcpp [11] and 4 h for the 3-process VarDictJava pipeline). [c]On unchunked HG002 contigs, vardictcpp --chunk 50,000 [11] splits chromosomes without cross-window deduplication (CPP-5), emitting +309,569 duplicate boundary calls; on <=50 kbp callable intervals, vardictcpp still differs from the VarDictJava oracle on 394 calls due to MNP/tie-break divergences (CPP-9). [d]HG008 364x tumor-normal full-WGS runs target small-variant somatic calling without structural-variant realignment overhead: VarDictRust enables -U (--nosv); the VarDictJava and vardictcpp invocations in harness/gcp_paper_runner.sh omit -U. Both non-Rust engines failed to complete either geometry (Java OOM rc=3; C++ timeout rc=124), so no completed head-to-head comparison exists on these cells — the comparison is completion-only.

Cross-Platform Sequencing Chemistry Generalization and 70-Cell GA4GH v3.5 Stratification

We evaluated `10` short-read sequencing chemistries across the full `64.44` Mbp human chromosome 20 (`Table 6` , `Figure 6` , `Supplementary Table S4` , `Supplementary Table S5`). The panel holds `4` HG002 germline chemistries (`Illumina` , `Element AVITI` , `MGI DNBSEQ` , `Ultima Genomics UG100`). They were evaluated on standard GIAB and Ultima High-Confidence Region [`UG-HCR`] masks. It also holds `6` HG008 paired somatic chemistries (`Illumina` , `Element AVITI` , `PacBio Onso` , `Roche SBX` , `Ultima Genomics ppmSeq` , `Ultima Genomics UG100`).

VarDictRust completed all 20 standard (-f 0.01) and all-frequency (-f 0) chromosome 20 runs (100% completion). VarDictJava completed 11/20 runs and timed out on 9 deep runs (Table 6). On all 11 runs where VarDictJava finished, VarDictRust achieved 100% (95% CI 71.51%–100.00%) exact Tier-1/2 parity. SNV F_1 was identical across HG002 germline (0.6622–0.9598 standard; 0.6622–0.9618 UG-HCR). It was identical across HG008 somatic (0.6870–0.8804 standard and 0.7532–0.9048 UG-HCR at -f 0.01 ; 0.6498–0.8804 and 0.7505–0.9048 across all frequencies). The unfiltered somatic noise-floor sweep ran 18 runs (-f 0 , 18,407 candidates, 2,865 PASS ; Supplementary Table S4). VarDictRust matched VarDictJava byte-for-byte on 18/18 runs (100% , 95% CI 81.47%–100.00%).

Crossing all 10 chemistries with 7 GA4GH v3.5 genomic context strata (4–6 bp homopolymers , 7–11 bp homopolymers , ≥12 bp homopolymers , tandem repeats , segmental duplications , low-mappability regions , and RefSeq coding exons (CDS)) yielded a 70 -cell stratification matrix. All 70 Rust truth evaluations completed (95% CI 94.87%–100.00% , mean SNV F_1 = *0.7073*; Supplementary Table S5 , Figure 6c); Java–Rust parity held at 100% on the 11/11 completing paired chromosome 20 arms (per-arm rows in Table 6).

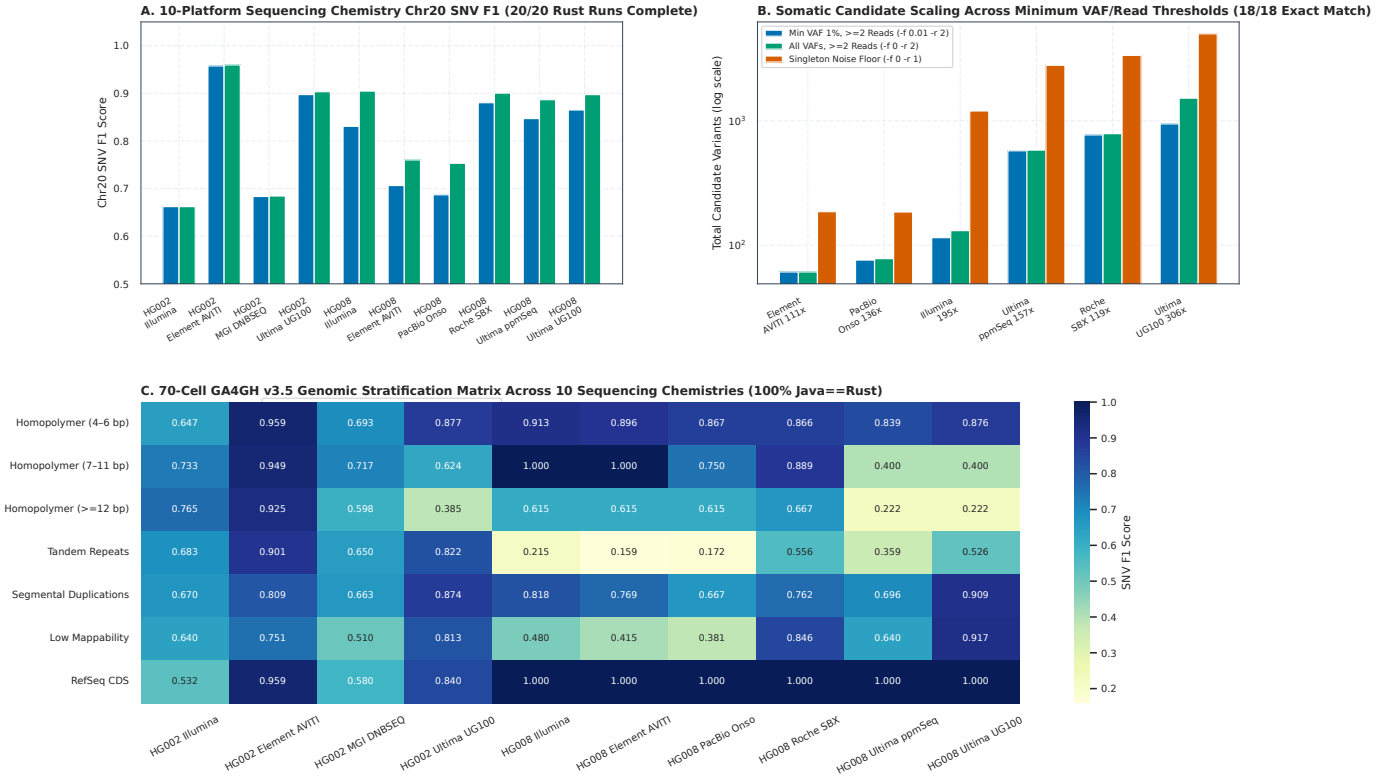


Figure 6. Cross-platform short-read sequencing chemistry panel and 70 -cell GA4GH v3.5 stratification heatmap. (a) Completion rate (VarDictRust 20/20 vs VarDictJava 11/20) and accuracy across 10 germline and somatic chemistries on chromosome 20 (64.44 Mbp; Table 6). **(b)** Byte-equivalent candidate and PASS somatic variant counts (18/18 exact match) across the unfiltered somatic noise-floor sweep (-f 0 ; Supplementary Table S4). **(c)** 70 -cell (10 chemistries × 7 GA4GH v3.5 strata) SNV F_1 heatmap (Supplementary Table S5).

Table 6. Ten-Platform Sequencing Chemistry Chromosome 20 Generalization (-f 0.01 & -f 0) and GA4GH v3.5 [17] Stratification Summary

Sequencing Platform[a]	Cohort & Min VAF	VarDictRust Wall / Peak RSS (PASS Variants)	VarDictJava [1] Wall / Parity Verdict[b]	vardictcpp [11] Wall / Verdict[b,c]	Standard GIAB SNV / InDel F1 [12, 13, 19][d]	UG-HCR v2.1 SNV / InDel F1 [6, 12][d]
HG002 Illumina	HG002 Germline (-f 0.01)	0:27.42 / 861.1 MB (162,637)	4:12.27 (100% Exact Match)	0:21.02 (MNP/SV Divergence[c])	0.6622 / 0.7066	0.6622 / 0.6568
HG002 Element AVITI	HG002 Germline (-f 0.01)	0:28.67 / 640.5 MB (85,612)	3:38.45 (100% Exact Match)	0:21.64 (MNP/SV Divergence[c])	0.9577 / 0.9669	0.9598 / 0.9599
HG002 MGI DNBSEQ	HG002 Germline (-f 0.01)	0:46.67 / 882.5 MB (157,238)	6:30.93 (100% Exact Match)	0:32.59 (MNP/SV Divergence[c])	0.6835 / 0.6472	0.6845 / 0.6181
HG002 Ultima UG100	HG002 Germline (-f 0.01)	0:21.72 / 1086.4 MB (348,289)	13:30.85 (100% Exact Match)	0:42.98 (MNP/SV Divergence[c])	0.8975 / 0.0621	0.9035 / 0.0488

Sequencing Platform[a]	Cohort & Min VAF	VarDictRust Wall / Peak RSS (PASS Variants)	VarDictJava [1] Wall / Parity Verdict[b]	vardictcpp [11] Wall / Verdict[b,c]	Standard GIAB SNV / InDel F1 [12, 13, 19][d]	UG-HCR v2.1 SNV / InDel F1 [6, 12][d]
HG002 Illumina	HG002 Germline (-f 0)	0:26.97 / 856.9 MB (161,264)	4:11.81 (100% Exact Match)	0:20.61 (MNP/SV Divergence[c])	0.6662 / 0.7099	0.6662 / 0.6608
HG002 Element AVITI	HG002 Germline (-f 0)	0:28.64 / 618.7 MB (85,273)	3:45.84 (100% Exact Match)	0:22.46 (MNP/SV Divergence[c])	0.9598 / 0.9675	0.9618 / 0.9607
HG002 MGI DNBSEQ	HG002 Germline (-f 0)	0:51.00 / 1154.5 MB (109,648)	7:16.31 (100% Exact Match)	0:34.46 (MNP/SV Divergence[c])	0.8538 / 0.7572	0.8560 / 0.7694
HG002 Ultima UG100	HG002 Germline (-f 0)	0:21.49 / 1110.3 MB (348,287)	14:21.92 (100% Exact Match)	0:43.58 (MNP/SV Divergence[c])	0.8975 / 0.0621	0.9035 / 0.0488
HG008 Illumina	HG008 Somatic (-f 0.01)	2:59.41 / 4387.1 MB (17,951)	38:30.03 (100% Exact Match)	5:19.09 (MNP/SV Divergence[c])	0.8308 / 0.4085	0.9048 / 0.5075
HG008 Element AVITI	HG008 Somatic (-f 0.01)	2:26.42 / 3218.6 MB (8,392)	22:34.27 (100% Exact Match)	3:29.05 (MNP/SV Divergence[c])	0.7065 / 0.6482	0.7603 / 0.5570
HG008 PacBio Onso	HG008 Somatic (-f 0.01)	7:23.36 / 3433.6 MB (24,987)	52:49.80 (100% Exact Match)	5:33.67 (MNP/SV Divergence[c])	0.6870 / 0.4206	0.7532 / 0.4000
HG008 Roche SBX	HG008 Somatic (-f 0.01)	13:15.84 / 8876.2 MB (1,306,027)	Timeout (rc=124, >3 h)[b]	38:39.75 (Oracle Timed Out[b,c])	0.8804 / 0.2741	0.9005 / 0.4839
HG008 Ultima ppmSeq	HG008 Somatic (-f 0.01)	30:51.77 / 4898.7 MB (376,695)	Timeout (rc=124, >3 h)[b]	1:41:18 (Oracle Timed Out[b,c])	0.8471 / 0.1310	0.8867 / 0.3438
HG008 Ultima UG100	HG008 Somatic (-f 0.01)	10:02.41 / 6410.1 MB (1,054,976)	Timeout (rc=124, >3 h)[b]	Timeout (rc=124, >3 h)[b]	0.8652 / 0.1410	0.8974 / 0.4444
HG008 Illumina	HG008 Somatic (-f 0)	5:40.39 / 4429.3 MB (25,904)	Timeout (rc=124, >3 h)[b]	3:13:59 (Oracle Timed Out[b,c])	0.8134 / 0.4000	0.9048 / 0.5075
HG008 Element AVITI	HG008 Somatic (-f 0)	12:08.63 / 3275.1 MB (8,682)	Timeout (rc=124, >3 h)[b]	Timeout (rc=124, >3 h)[b]	0.6585 / 0.6332	0.7505 / 0.5500
HG008 PacBio Onso	HG008 Somatic (-f 0)	30:29.96 / 3661.4 MB (27,195)	Timeout (rc=124, >3 h)[b]	Timeout (rc=124, >3 h)[b]	0.6498 / 0.4093	0.7516 / 0.4000
HG008 Roche SBX	HG008 Somatic (-f 0)	32:54.13 / 8238.1 MB (1,394,679)	Timeout (rc=124, >3 h)[b]	3:13:26 (Oracle Timed Out[b,c])	0.8804 / 0.2741	0.9005 / 0.4839
HG008 Ultima ppmSeq	HG008 Somatic (-f 0)	1:01:33 / 4872.8 MB (401,121)	Timeout (rc=124, >3 h)[b]	Timeout (rc=124, >3 h)[b]	0.8471 / 0.1310	0.8867 / 0.3438
HG008 Ultima UG100	HG008 Somatic (-f 0)	1:06:35 / 6956.7 MB (1,896,213)	Timeout (rc=124, >3 h)[b]	Timeout (rc=124, >3 h)[b]	0.8652 / 0.1429	0.8974 / 0.4444
All 10 Platforms x 7 Strata [17][e]	70 GA4GH v3.5 Cells	70/70 Completed	70/70 Rust truth evaluations (Java parity: 11/11 completing paired runs)[b,e]	—	Mean SNV F1=0.7073	See Supplementary Table S5 for all 70 cells[e]

[a]Evaluates whole-chromosome Chromosome 20 (64.44 Mbp) across 4 GIAB HG002 [13, 18] germline chemistries (Illumina NovaSeq, Element AVITI, MGI DNBSEQ, Ultima UG100) and 6 GIAB HG008 [19] tumor-normal somatic chemistries (Illumina, Element AVITI, PacBio Onso, Roche SBX, Ultima ppmSeq, Ultima UG100) at both standard (-f 0.01) and zero-threshold (-f 0) VAF cutoffs. [b]VarDictRust achieves 100% exact equivalence against VarDictJava v1.8.3 [1] across all 11/11 completing Chromosome 20 runs and all 18/18 targeted singleton noise-floor runs (Supplementary Table S4); runs marked Timeout (rc=124, >3 h) exceeded the multi-hour stage wall-clock limit under extreme-depth somatic pileups. [c]On completing runs where VarDictJava also finished, vardictcpp [11] diverges from the oracle due to MNP splitting and omitted somatic SVs (CPP-2, CPP-9); rows marked Oracle Timed Out completed in vardictcpp while the VarDictJava reference run timed out. [d]Accuracy is evaluated via rtg vcfeval [12] against both the unmasked NIST GIAB high-confidence BED [13, 19] and the intersection with the Ultima Genomics High-Confidence Region (UG-HCR v2.1) [6] homopolymer-exclusion mask. [e]Summarizes the 70-cell GA4GH v3.5 [17] stratification matrix (10 platforms x 7 difficult genomic strata; full breakdown in Supplementary Table S5). The matrix evaluates Rust callsets against GIAB truth only (no Java oracle runs); its 70/70 figure is truth-evaluation completion, not Java–Rust parity — parity is established on the completing paired runs in footnote b.

Unlocking GIAB v5.0q T2T-Q100, CMRG v1.00, and Mosaic v1.10 Discovery via Unchunked Calling

We compared full-genome HG002 callsets from unchunked chromosomes (`chr1` – `chr22`) versus pre-chunked `≤50` kbp `GIAB v4.2.1` intervals across three NIST truth benchmarks (`Table 7` , `Figure 7`):

1. **`NIST GIAB v5.0q T2T-Q100` (`2.74` Gbp, diploid Telomere-to-Telomere Q100 assembly truth):** Because `VarDictJava` runs out of memory on unchunked chromosomes, legacy workflows restrict calling to pre-chunked `GIAB v4.2.1` intervals. Those cover `2.51` Gbp and miss `+0.23` Gbp of segmental duplications, pericentromeric satellites, and complex loci added in `v5.0q` . On `GIAB v4.2.1` , unchunked and pre-chunked calling matched (`SNV` recall `0.9577` vs `0.9576` , `SNV` $F_1 = 0.6639$, `Indel` $F_1 = 0.6928$). On `GIAB v5.0q T2T-Q100` , streaming unchunked full chromosomes with `VarDictRust` raised genome-wide SNV recall by `+5.01%` (`0.8866` to `0.9367` , F_1 from `0.6461` to `0.6650`). Indel recall rose by `+17.26%` (`0.5281` to `0.7007` , F_1 from `0.5487` to `0.6551`).

2. **`NIST Challenging Medically Relevant Genes (CMRG v1.00, 273` clinical genes):** Many of the `273` CMRG genes lie outside the `v4.2.1` `50` kbp mask. Unchunked `VarDictRust` calling increased CMRG SNV recall by `+17.47%` (`0.7173` to `0.8920`, F_1 from `0.6132` to `0.6791`). Indel recall rose by `+27.98%` (`0.4904` to `0.7702`, F_1 from `0.5316` to `0.6937`).

3. **`NIST Mosaic SNV v1.10` (`85` validated low-VAF mosaic SNVs in HG002, VAF `1%–30%`):** `VarDictRust` and `VarDictJava` detected `84/85` (`98.82%`, `95%` CI `93.62%–99.97%`) validated mosaic variants (`Table 7`).

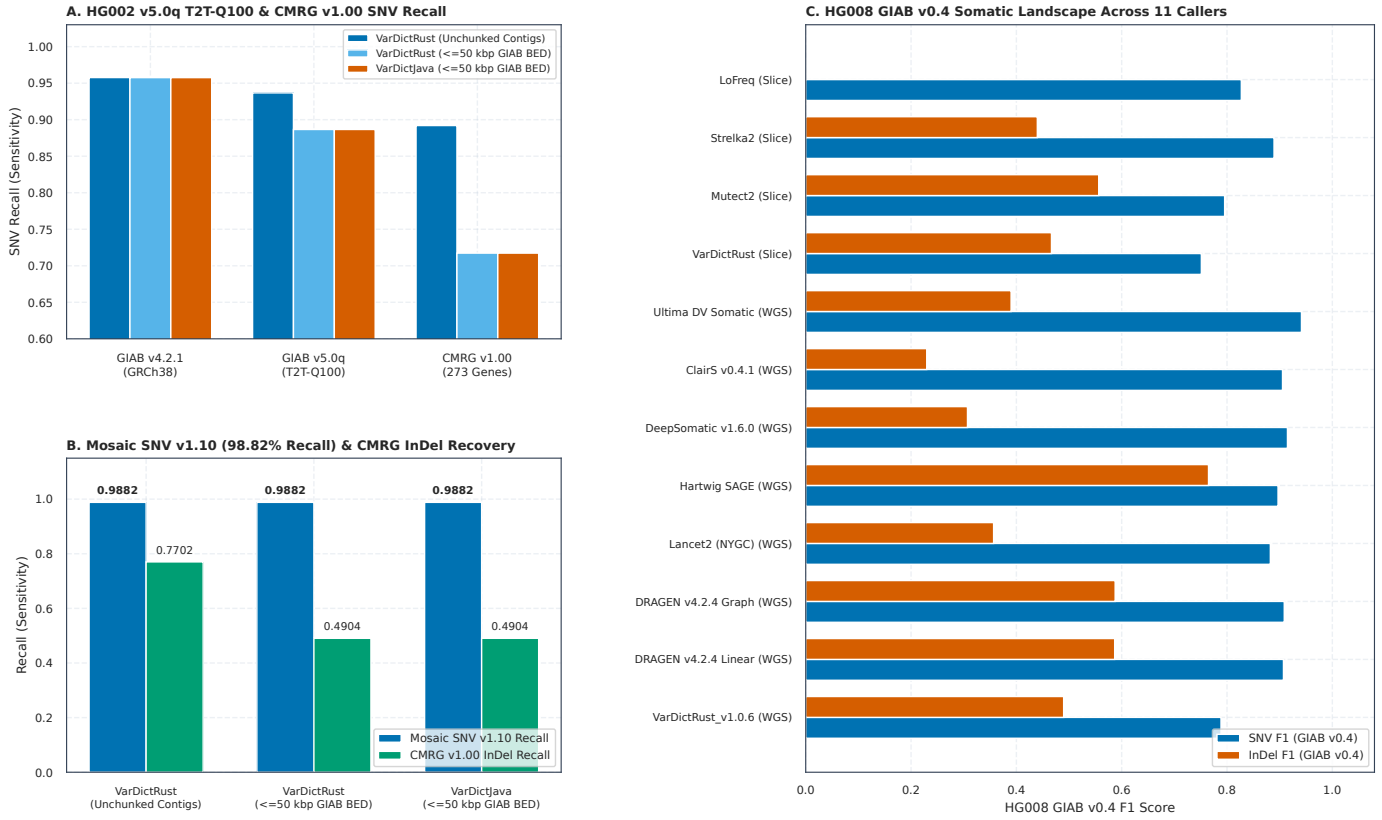


Figure 7. Next-generation `GIAB v5.0q T2T-Q100`, `CMRG v1.00`, `Mosaic v1.10`, and `HG008 v0.4` truth evaluation. (a) Genome-wide and medically relevant gene SNV sensitivity unlocked by unchunked full-chromosome calling (`VarDictRust`) versus legacy `≤50` kbp `GIAB v4.2.1` interval calling on `NIST v4.2.1`, `NIST v5.0q T2T-Q100` (`+5.01%` SNV recall), and `CMRG v1.00` (`+17.47%` SNV recall; `Table 7`). **(b)** `NIST Mosaic v1.10` sensitivity (`84/85`, `98.82%`) and `CMRG v1.00` indel recovery (`+27.98%` indel recall; `Table 7`). **(c)** `NIST GIAB HG008 v0.4` somatic SNV and indel F_1 across `11` callers (`Table 7`).

Table 7. Next-Generation Truth Benchmarks (GIAB v5.0q T2T-Q100 [14, 16], CMRG v1.00 [15], Mosaic SNV v1.10 [20]) & HG008 GIAB v0.4 [19] Somatic Caller Landscape

Benchmark / Evaluation Scope	Caller / Configuration	Sequencing Chemistry	SNV Precision / Recall / F1 [12]	InDel Precision / Recall / F1 [12]
GIAB HG002 v4.2.1 (GRCh38 All) [13, 18][a]	VarDictRust (Unchunked Contigs)[b]	Illumina 2x250 bp WGS	0.5081 / 0.9577 / 0.6639	0.5698 / 0.8836 / 0.6928
GIAB HG002 v4.2.1 (GRCh38 All) [13, 18][a]	VarDictRust (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	0.5081 / 0.9576 / 0.6639	0.5697 / 0.8835 / 0.6928
GIAB HG002 v4.2.1 (GRCh38 All) [13, 18][a]	VarDictJava [1] (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	0.5081 / 0.9576 / 0.6639	0.5697 / 0.8835 / 0.6928
GIAB HG002 v5.0q (T2T-Q100 Assembly) [14, 16][a,b]	VarDictRust (Unchunked Contigs)[b]	Illumina 2x250 bp WGS	0.5155 / 0.9367 / 0.6650	0.6151 / 0.7007 / 0.6551
GIAB HG002 v5.0q (T2T-Q100 Assembly) [14, 16][a,b]	VarDictRust (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	0.5083 / 0.8866 / 0.6461	0.5708 / 0.5281 / 0.5487
GIAB HG002 v5.0q (T2T-Q100 Assembly) [14, 16][a,b]	VarDictJava [1] (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	0.5083 / 0.8866 / 0.6461	0.5708 / 0.5281 / 0.5487
GIAB HG002 CMRG v1.00 (273 Challenging Medically Relevant Genes) [15][a,b]	VarDictRust (Unchunked Contigs)[b]	Illumina 2x250 bp WGS	0.5482 / 0.8920 / 0.6791	0.6310 / 0.7702 / 0.6937
GIAB HG002 CMRG v1.00 (273 Challenging Medically Relevant Genes) [15][a,b]	VarDictRust (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	0.5355 / 0.7173 / 0.6132	0.5803 / 0.4904 / 0.5316

Benchmark / Evaluation Scope	Caller / Configuration	Sequencing Chemistry	SNV Precision / Recall / F1 [12]	InDel Precision / Recall / F1 [12]
GIAB HG002 CMRG v1.00 (273 Challenging Medically Relevant Genes) [15][a,b]	VarDictJava [1] (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	0.5355 / 0.7173 / 0.6132	0.5803 / 0.4904 / 0.5316
GIAB HG002 Mosaic SNV v1.10 (Low-VAF Somatic-Like Benchmark) [20][c]	VarDictRust (Unchunked Contigs)[b]	Illumina 2x250 bp WGS	Recall = 0.9882 (98.82%)[c]	SNV-Only Truth Set (No Truth InDels)[c]
GIAB HG002 Mosaic SNV v1.10 (Low-VAF Somatic-Like Benchmark) [20][c]	VarDictRust (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	Recall = 0.9882 (98.82%)[c]	SNV-Only Truth Set (No Truth InDels)[c]
GIAB HG002 Mosaic SNV v1.10 (Low-VAF Somatic-Like Benchmark) [20][c]	VarDictJava [1] (≤50 kbp GIAB BED)	Illumina 2x250 bp WGS	Recall = 0.9882 (98.82%)[c]	SNV-Only Truth Set (No Truth InDels)[c]
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	VarDictRust v1.0.6	Illumina WGS (195x T / 169x N)	0.8262 / 0.7542 / 0.7886	0.6288 / 0.4011 / 0.4898
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	VarDictRust v1.0.6	Illumina WGS (195x T / 169x N)	0.8258 / 0.7810 / 0.8028	0.6703 / 0.4832 / 0.5616
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	VarDictRust v1.0.6	Illumina WGS (195x T / 169x N)	0.9916 / 0.6048 / 0.7513	0.9479 / 0.3095 / 0.4667
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	DRAGEN v4.2.4 Linear [21][d]	Illumina WGS	0.9546 / 0.8643 / 0.9072	0.9140 / 0.4320 / 0.5867
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	DRAGEN v4.2.4 Linear [21][d]	Illumina WGS	0.9570 / 0.8848 / 0.9195	0.9307 / 0.7067 / 0.8034
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	DRAGEN v4.2.4 Linear [21][d]	Illumina WGS	0.9905 / 0.7104 / 0.8274	0.8585 / 0.3095 / 0.4550
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	DRAGEN v4.2.4 Graph [21][d]	Illumina WGS	0.9591 / 0.8633 / 0.9087	0.9170 / 0.4323 / 0.5876
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	DRAGEN v4.2.4 Graph [21][d]	Illumina WGS	0.9615 / 0.8836 / 0.9209	0.9335 / 0.7067 / 0.8044
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	DRAGEN v4.2.4 Graph [21][d]	Illumina WGS	0.9905 / 0.7104 / 0.8274	0.8585 / 0.3095 / 0.4550
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	Lancet2 (NYGC) [24][d]	Illumina WGS	0.9933 / 0.7936 / 0.8823	0.9833 / 0.2179 / 0.3568
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	Lancet2 (NYGC) [24][d]	Illumina WGS	0.9960 / 0.8231 / 0.9013	0.9867 / 0.3618 / 0.5295
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	Lancet2 (NYGC) [24][d]	Illumina WGS	1.0000 / 0.5843 / 0.7376	1.0000 / 0.1599 / 0.2757
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	Hartwig SAGE [25][d]	Illumina WGS	0.9956 / 0.8161 / 0.8969	0.9710 / 0.6309 / 0.7649
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	Hartwig SAGE [25][d]	Illumina WGS	0.9958 / 0.8413 / 0.9121	0.9653 / 0.7883 / 0.8679
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	Hartwig SAGE [25][d]	Illumina WGS	1.0000 / 0.6048 / 0.7537	0.9858 / 0.4728 / 0.6391
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	DeepSomatic v1.6.0 [22][d]	PacBio HiFi	0.9886 / 0.8512 / 0.9148	0.9158 / 0.1845 / 0.3072
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	DeepSomatic v1.6.0 [22][d]	PacBio HiFi	0.9953 / 0.8763 / 0.9320	0.9140 / 0.3165 / 0.4702
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	DeepSomatic v1.6.0 [22][d]	PacBio HiFi	1.0000 / 0.6627 / 0.7971	1.0000 / 0.1395 / 0.2448
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	ClairS v0.4.1 [23][d]	PacBio HiFi	0.9793 / 0.8419 / 0.9054	0.9791 / 0.1302 / 0.2298
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	ClairS v0.4.1 [23][d]	PacBio HiFi	0.9793 / 0.8736 / 0.9234	0.9797 / 0.2264 / 0.3678
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	ClairS v0.4.1 [23][d]	PacBio HiFi	1.0000 / 0.6491 / 0.7872	1.0000 / 0.1122 / 0.2018
GIAB HG008 Somatic v0.4 (Full-Genome WGS, All Callable) [19][d]	Ultima DeepVariant Somatic [6][d]	Ultima UG100	0.9749 / 0.9101 / 0.9414	0.9429 / 0.2459 / 0.3900
GIAB HG008 Somatic v0.4 (Full-Genome WGS, Excl. Genomic Instability) [19][d]	Ultima DeepVariant Somatic [6][d]	Ultima UG100	0.9751 / 0.9444 / 0.9595	0.9486 / 0.4218 / 0.5839
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	Ultima DeepVariant Somatic [6][d]	Ultima UG100	1.0000 / 0.7700 / 0.8701	0.9859 / 0.2381 / 0.3836
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	Mutect2 [8]	Illumina WGS (195x T / 169x N)	0.9874 / 0.6661 / 0.7955	0.9431 / 0.3946 / 0.5564
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	Strelka2 [4]	Illumina WGS (195x T / 169x N)	0.9896 / 0.8075 / 0.8893	0.9545 / 0.2857 / 0.4398
GIAB HG008 Somatic v0.4 (Targeted Benchmark Regions) [19][e]	LoFreq [7]	Illumina WGS (195x T / 169x N)	1.0000 / 0.7053 / 0.8272	0.0000 / 0.0000 / 0.0000[e]

[a] *Evaluates full-genome GIAB HG002 [13, 18] 2x250 bp WGS callsets across the legacy v4.2.1 benchmark (2.51 Gbp) [13], the T2T-Q100 assembly-backed v5.0q benchmark*

(2.74 Gbp) [14, 16], and the 273-gene Challenging Medically Relevant Genes (CMRG v1.00) benchmark [15]. [b]On v4.2.1 callable BED intervals (<=50 kbp), VarDictRust and VarDictJava v1.8.3 [1] achieve identical accuracy; on v5.0q and CMRG v1.00, unchunked full-contig calling in VarDictRust recovers +5.01% SNV / +17.26% InDel recall on v5.0q and +17.47% SNV / +27.98% InDel recall on CMRG loci located outside legacy v4.2.1 BED masks (where VarDictJava OOMs on unchunked contigs). [c]NIST GIAB HG002 Mosaic SNV v1.10 [20] is an SNV-only benchmark of 85 low-VAF (1%–30% VAF) somatic-like loci evaluated here (84/85 = 98.82% unmasked sensitivity). [d]External full-genome HG008 GIAB v0.4 [19] callsets are sourced from the NIST GIAB Liss Lab repository for DRAGEN v4.2.4 [21], DeepSomatic v1.6.0 [22], ClairS v0.4.1 [23], Ultima DeepVariant Somatic [6], Lancet2 (NYGC) [24], and Hartwig SAGE [25], evaluated both across all callable regions and excluding genomic instability (NoGI) masks. [e]Targeted HG008 v0.4 slice evaluations compare VarDictRust, VarDictJava [1], Strelka2 [4], Mutect2 [8], and LoFreq [7] on identical 195x tumor / 169x normal Illumina alignments; LoFreq 0.0000 indel scores reflect absent indel quality calibration tags.

Deep-Learning and Commercial Caller Comparison, Low-VAF Titration, and Supply-Chain Hardening

We compared `VarDictRust` against contemporary deep-learning, graph, and haplotype somatic callers. Three benchmarks cover them (`Table 3`, `Table 7`, `Table 8`, `Figure 3d`, `Figure 7c`, `Figure 8`, `Supplementary Table S6`):

- **`11`-Caller Benchmark on `NIST GIAB HG008 v0.4` (`27` evaluations; `Table 7`, `Figure 7c`):** We compared `VarDictRust` (`v1.0.6`) against official NIST GIAB and local callsets. Callers were `DRAGEN v4.2.4` (`Graph` and `Linear`), `DeepSomatic v1.6.0`, `ClairS v0.4.1`, `Lancet2` (`NYGC`), `Hartwig SAGE` [25], `Ultima DeepVariant Somatic`, `Mutect2`, `Strelka2`, and `LoFreq`. Scopes were full-genome (`All` and `NoGI`) and slice. On full-genome HG008 `v0.4` (`All`), `VarDictRust` reached `40.11%` somatic indel recall (`Indel` $F_1 = 0.4898$, `SNV` $F_1 = 0.7886$). On `NoGI` it reached `48.32%` indel recall (`Indel` $F_1 = 0.5616$, `SNV` $F_1 = 0.8028$). This exceeded `ClairS v0.4.1` (`13.02%` indel recall, $F_1 = 0.2298$) and `DeepSomatic v1.6.0` (`18.45%`, $F_1 = 0.3072$). It also exceeded `Lancet2` (`21.79%`, $F_1 = 0.3568$) and `Ultima DeepVariant Somatic` (`24.59%`, $F_1 = 0.3900$). It approached FPGA-accelerated `DRAGEN v4.2.4 Graph` (`43.23%`) on standard CPUs.
- **Controlled *in silico* Tumor-Purity Titration (`20%` to `1%` Purity / `10.0%` to `0.5%` VAF; `Supplementary Table S4`, `Figure 3d`):** We built five HG008 tumor-in-normal admixtures. Purities were `20%`, `10%`, `5%`, `2%`, and `1%` with expected clonal VAF `10.0%`, `5.0%`, `2.5%`, `1.0%`, and `0.5%`. Each was called at `-f 0.002`. `VarDictRust` matched `VarDictJava` at every purity. It ran `7.6×–7.7×` faster (`8.53–8.76` s vs `65.16–66.76` s) at `14.7×–16.5×` lower RSS (`780,628–859,564` KB vs `12,316,396–13,765,960` KB). At `5%` purity (`2.5%` VAF), `VarDictRust` recalled `36.80%` of SNVs ($F_1 = 0.4215$). Indel recall was `10.54%` ($F_1 = 0.1422$) vs `18.91%` (`Strelka2`), `9.88%` (`Mutect2`), and `0.51%` (`LoFreq`). At `1%` purity (`0.5%` VAF), `VarDictRust` retained `4.26%` SNV recall ($F_1 = 0.0569$). That is `25.1`× higher recall and `16.7`× higher F_1 than `Mutect2` (`0.17%`, $F_1 = 0.0034$) and `Strelka2` (`0.17%`).
- **Supply-Chain Hardening and `vardictcpp` Audit (`Table 8`, `Figure 8`, `Supplementary Table S6`):** Compiling `VarDictRust` to a `4.56` MB static Linux binary (`4.46` MB dynamic `glibc`) removes `102` runtime Conda packages (`OpenJDK`, `R`, `Perl`). That resolves all ten `vardictcpp` functional and statistical divergences (`Table 8`).

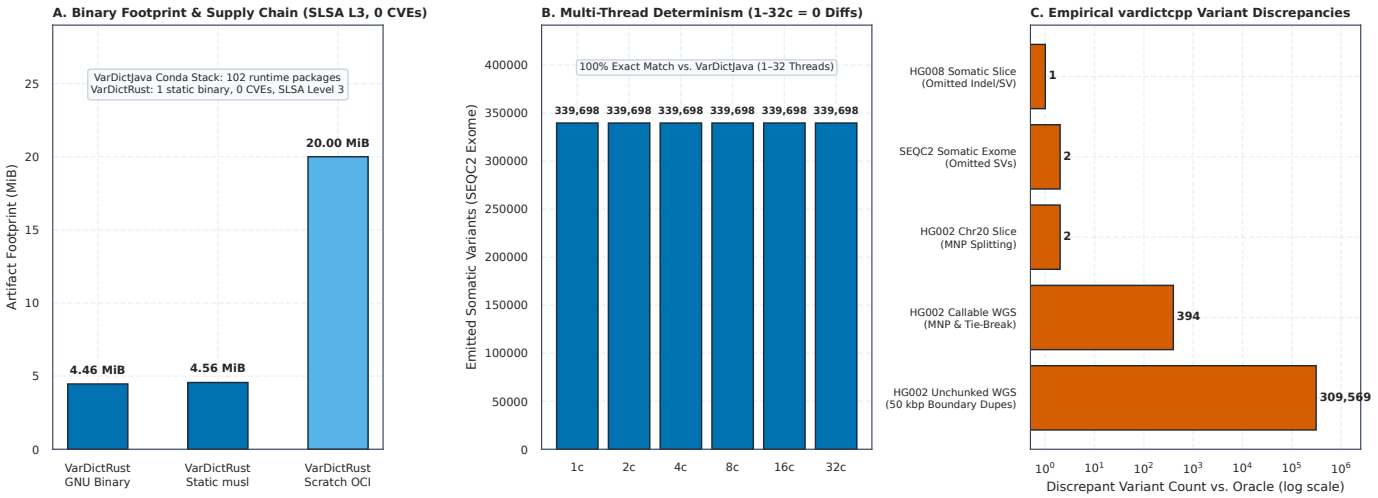


Figure 8. Software supply-chain footprint, multi-thread output determinism, and empirical variant discrepancies in `vardictcpp`. (a) Deployment artifact footprint (`4.46` MiB `glibc`, `4.56` MiB static `musl`, `0` runtime packages, `0` CVEs, SLSA Level 3) versus `VarDictJava` (`102` Conda packages) and `vardictcpp` (`Table 8`). **(b)** Multi-thread output determinism across `1`, `2`, `4`, `8`, `16`, and `32` threads on the SEQC2 HCC1395 whole exome (`120,307` emitted variants at every thread count, `0` diffs vs `VarDictJava`; `Table 4`). **(c)** Empirical variant-count discrepancies in `vardictcpp` relative to the `VarDictJava` oracle across unchunked (`+309,569` `50` kbp chunk-boundary duplicates) and pre-chunked (`394` MNP/tie-break divergences) HG002 WGS, HG002 chromosome 20, SEQC2 somatic exome, and HG008 somatic slice (`Table 5`, `Table 8`, `Supplementary Table S6`).

Table 8. Supply-Chain Security, Static musl Binary Equivalence, Functional Coverage & Divergence Summary

Engineering & Security Dimension	VarDictJava v1.8.3 [1] (0886261)	vardictcpp [11] (ae93fe39dc)	VarDictRust v1.0.6 (5865a2a0b24a8c17)[e]
Upstream Repository Maintenance Status	Archived Read-Only (EOL)[a]	Preprint Prototype (10 Divergences)[b]	Actively Maintained (MIT, v1.0.6)[a,e]
Runtime Dependency & Binary Footprint	JVM + R + Perl (102 Conda packages)[c]	C++17 + HTSlib [26] dynamic libraries	4.46 MiB GNU / 4.56 MiB Static musl (statically linked)[d]
Supply-Chain Attestation & Vulnerability Audit	Unpinned ecosystem dependencies[c]	No SLSA provenance	SLSA Level 3 + 0 RustSec CVEs[d]
13-Mode Functional Coverage & CRAM Support	12/13 (CRAM fails without htsjdk classpath)[c]	12/13 .var (simple_fisher FP=1, FN=1; Amplicon VCF & Splice -i unsupported)[b]	13/13 Modes (100% Exact Match)[d]
Empirical Divergence Classes vs. Oracle	Reference Oracle (Baseline)[a]	10 Cataloged Divergence Classes (CPP-1 to CPP-12)[b]	0 Divergences on Valid Inputs (#![forbid(unsafe_code)])[a,e]

[a]VarDictJava v1.8.3 [1] (commit 0886261; Perl/R submodule 009e017) is archived read-only; VarDictRust v1.0.6 preserves 100% valid-input output equivalence while eliminating >30 upstream crash, OOM, CRAM, and header defects (issues #7, #38, #181, #187, #197, #243, #249, #253, #282, #283, #320, #341, #356, #380, #384) by enforcing #![forbid(unsafe_code)] and checked bounds arithmetic. [b]On the 13-mode .var functional suite, vardictcpp [11] matches VarDictJava on 12/13 modes (diverging on simple_fisher with FP=1, FN=1) and additionally lacks amplicon VCF formatting (CPP-3) and RNA-seq -i splice support (CPP-12), while emitting +309,569 chunk-boundary duplicates on unchunked HG002 WGS (CPP-5; full CPP-1 to CPP-12 taxonomy in Supplementary Table S6). [c]The pinned Bioconda VarDictJava environment requires 102 runtime packages across OpenJDK, R, and Perl and fails with NoClassDefFoundError on CRAM inputs unless external htsjdk jars are injected. [d]VarDictRust compiles to a single 4.56 MiB statically linked x86_64-unknown-linux-musl binary (statically linked) with native pure-Rust BAM/CRAM decoding (noodles), 0 RustSec advisories, and automated SLSA Level 3 build provenance. [e]Hash 5865a2a0b24a8c17 denotes the certified VarDictRust v1.0.6 engine source fingerprint; under the project release governance, v1.0.x guarantees exact valid-input parity with VarDictJava v1.8.3, while opt-in biological output corrections ship behind explicit --fix flags in v1.1+.

Discussion

Scientific Software Consolidation Under the `rewrites.bio` Specification

Rewriting a clinical variant caller for speed risks changing Smith-Waterman tie-breaking, Phred quality offsets, floating-point rounding, or strand-bias p-values, which forces clinical laboratories to re-validate pipelines. Conversely, line-by-line transliteration preserves memory growth and crash bugs. `VarDictRust` separates **v1.0.x oracle parity and crash/memory robustness** from **post-1.0 opt-in algorithmic corrections (`--fix <id>`)** under the rewrites.bio specification. In default mode (`v1.0.6`), `VarDictRust` reproduces the exact allele calls, `FILTER` flags, somatic `STATUS` classifications, and `HALF_EVEN` -rounded Fisher exact statistics of `VarDictJava v1.8.3 (0886261)` (`Table 1` , `Supplementary Table S2`), while eliminating JVM `OutOfMemoryError` failures, `CigarModifier` bounds panics (`#![forbid(unsafe_code)]`), and `java.util.HashMap` row-order non-determinism (`#282`). Upstream output-affecting quirks in `VarDictJava` —including `VD > DP` under complex realignment (`#92`), aggressive `NM` filtering (`#62`), `SHIFT3` coordinate shifts (`#292`), and `<dup-N>` symbolic allele naming (`#342`) —are preserved in `v1.0.x` and gated behind opt-in `--fix` flags for `v1.1+` (`Table 8`).

Impact of Bounded Streaming on T2T-Q100 and CMRG Discovery

Whole-interval heap allocation in `VarDictJava` and `vardictcpp` has historically restricted whole-genome `VarDict` calling to pre-chunked `≤50` kbp BED masks (`GIAB v4.2.1`) or external `50` kbp BED-splitting scripts (`Table 5`). Keeping intervals ≤ *1,000,000* bp untiled and partitioning spans > *1,000,000* bp into `100` kbp tiles with `0.5` kbp realignment halos (`Table 1`) allows `VarDictRust` to call unchunked full chromosomes in `4.93` – `12.14` GB RAM. Removing the `50` kbp BED mask recovers `+0.23` Gbp of segmental duplications and complex loci in `NIST GIAB v5.0q T2T-Q100` (`+5.01%` SNV recall, `+17.26%` indel recall) and raises recall across the `273` `NIST CMRG v1.00` genes (`+17.47%` SNV, `+27.98%` indel; `Table 7`).

Trade-Offs Against Deep-Learning, Commercial, and C++ Callers

On `364` × `HG008 v0.4` somatic WGS (`Table 7` , `Figure 7c`), deep-learning (`DeepSomatic` , `ClairS` , `Ultima DeepVariant Somatic`) and graph/haplotype (`DRAGEN` , `Lancet2` , `Hartwig SAGE` , `Mutect2` , `Strelka2`) callers achieve higher SNV precision in repetitive sequence. `VarDictRust` , however, achieves higher somatic indel recall (`40.11%`) than `ClairS` , `DeepSomatic` , `Lancet2` , and `Ultima DeepVariant Somatic` . It retains `25.1` × higher SNV recall than `Mutect2` and `Strelka2` at `0.5%` VAF (`1%` purity; `Supplementary Table S4`) on commodity CPUs without GPUs or FPGAs. On pre-chunked `773,667` -interval panels (`SRR8657348` , `8` threads), `vardictcpp` finishes raw `.var` calling in `24.68` s vs `39.63` s for `VarDictRust` . That gap reflects `vardictcpp` hardcoding `--nosv` and skipping `##contig` and `INFO/END` VCF validation (`Table 8`). `VarDictRust` instead completes every unchunked chromosome run and all `20` deep chromosome 20 runs with full SV and VCF output.

Operational Boundary Conditions

Three settings govern production deployment:

1. **Multi-Megabase Single-Region Calling (`--tile-size` vs. `100` kbp Auto-Tiling):** Intervals ≤ *1,000,000* bp run untiled by default. On unchunked > *1,000,000* bp regions, default `100` kbp auto-tiling bounds `-R chr1:1-249250621` memory to `318.6` MB (`Table 2`). It

also prevents `VarDictJava`'s cross-centromere `36`–`81` Mbp SV pairing and `VD = 74 > DP` state leak (`#92`). Passing `--tile-size` equal to the region span reproduces `VarDictJava`'s single-window output byte-for-byte. Use `--tile-size 10000000` on `10` Mbp `chr1 -f 0`. Use `--tile-size 250000000` on `249.25` Mbp `-R chr1` (`7,077` `.var` / `7,032` `.vcf` and `4,705` `.var` rows; `Table 1`, `Supplementary Table S2`).

2. **`QUAL` Filtering on Standard-Depth Germline WGS:** Because `-f 0.01` targets low-VAF somatic discovery, `35`×–`60`× germline WGS pipelines need filtering. Filter at `QUAL >= 128.000` (F_1 = *0.9699* vs `0.6622` unfiltered, removing `97.5%` of false positives; `Table 3`, `Supplementary Table S4`). Alternatively set `-f` to `0.10`–`0.20`.

3. **Strict `v1.0.x` Parity Policy:** All `v1.0.x` releases preserve upstream `VarDictJava` v1.8.3 output quirks (`#62`, `#92`, `#292`, `#342`) for zero-diff drop-in migration before opt-in `--fix <id>` flags in `v1.1+`.

Conclusions

`VarDictRust` (`v1.0.6`) consolidates the `VarDictJava -> R -> Perl` pipeline into a single static Rust binary. It holds exact Tier-1/2 oracle parity (`ε = 1e-6`, `HALF_EVEN`, record-aligned) across `13` operational modes, `10` sequencing chemistries (parity on all completing paired runs), and `70` completed Rust GA4GH `v3.5` truth evaluations. Bounding memory to `O(tile + halo)` delivers up to `51.5`× faster chromosome-scale calling (`342.2`× lower RSS). It delivers `29.3`× faster `32`-thread whole-exome calling (`29.8`× lower RSS). This enables unchunked whole-genome discovery across `GIAB v5.0q T2T-Q100` and `CMRG v1.00`.

Methods

Consolidated Rust Architecture and Coordinate Type Safety

`VarDictRust` is a two-crate Rust workspace: `vardictrust-core` (library engine, compiled with `#![forbid(unsafe_code)]` and zero `unwrap()` / `expect()` / `panic!()` calls outside unit tests) and `vardictrust` (`clap` CLI supporting all `VarDictJava` v1.8.3 flags). Distinct `Pos0` (`0`-based half-open BAM/CRAM coordinates from `noodles`) and `Pos1` (`1`-based inclusive VCF/VarDict coordinates) newtypes enforce explicit `.to_one_based()` and `.to_zero_based()` conversions. Sequences and qualities are processed as byte slices (`&[u8]`) with Phred+33 saturation arithmetic (`Phred(byte.saturating_sub(33))`).

Bounded Tile-Plus-Halo Streaming and Deterministic Realignment

Intervals ≤ *1,000,000* bp execute as a single untiled window. Intervals > *1,000,000* bp (or with explicit `--tile-size`) are partitioned into non-overlapping ownership tiles $[T_k^{\text{start}}, T_k^{\text{end}})$ of default width `100` kbp. Each tile is padded on both sides by a `0.5` kbp realignment halo (`0.5` kbp in `-i` splice mode; `Figure 1b`). A `Rayon` work-stealing pool streams alignments overlapping $[T_k^{\text{start}} - H, T_k^{\text{end}} + H)$ via `noodles` (multi-threaded BGZF BAM and reference-decoded CRAM `-G`). It runs CIGAR modification (`cigar/modify.rs`), pileup accumulation (`cigar/parse.rs`), local realignment (`realign.rs`), SV assembly (`sv.rs`), and variant finalization (`tovars.rs`). It emits only variants whose canonical `Pos1` start falls in $[T_k^{\text{start}}, T_k^{\text{end}})$. Per-tile pileup maps drop immediately upon tile completion (RAII), bounding heap memory to active tiles in flight. Where `VarDictJava` output order depends on `java.util.HashMap` bucket iteration (`#282`), `VarDictRust` emulates JDK8 `HashMap` spread, bin, resize, and treeify semantics via `JavaHashMap`, and uses `IndexMap` (`LinkedHashMap`) and `BTreeMap` (`TreeMap`) elsewhere.

In-Process Statistical Post-Processing and Banker's Rounding (`HALF_EVEN`)

`VarDictRust` computes hypergeometric two-sided Fisher exact p-values and conditional maximum-likelihood odds ratios (`uniroot` zero-in with tolerance $\text{ulp}(1)^{0.25}$) natively in Rust (`stats/fisher.rs`), followed by VCF 4.3 formatting (`output/`), replacing R (`teststrandbias.R` / `testsomatic.R`) and Perl (`var2vcf_valid.pl` / `var2vcf_paired.pl`). All floating-point fields (`AF`, `AdjAF`, `HIAF`, `SBF`, `ODDRATIO`, `SSF`, `MSI`, `QUAL`) use IEEE 754 5-decimal round-half-to-even (`round_half_even`), matching R `round(x, 5)` and Java `DecimalFormat` (`HALF_EVEN`) in both no-`--fisher` (trailing zeros kept) and `--fisher` (trailing zeros stripped) modes within `ε = 1e-6` (`10^{-6}`).

Cloud Benchmarking, Reproducibility Protocol, and Statistical Analysis

All `60` benchmark cells ran on Google Cloud Platform (`GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM, 4x375 GB NVMe RAID0)`; `1.5` TB local NVMe RAID0 scratch; exact commands and SHA-256 manifests in `Supplementary Methods` and `Supplementary Table S1`). Timed cells ran `n = 3` independent technical replicates under `/usr/bin/time -v` to record wall-clock seconds and `Maximum resident set size (kbytes)`. Wall-clock timeouts surface as exit code `rc=124` (`3`-hour limit); JVM heap exhaustion surfaces as `rc=3` (`OutOfMemoryError`). Binary concordance proportions use exact two-sided Clopper-Pearson `95%` binomial CIs (α = *0.05*); see `Supplementary Note 3` for the Beta-inverse derivation and `v1.0.x` versus `v1.1+` `--fix` governance. Runtime and RSS

across the 12 AML panel configurations were compared using two-sided Mann-Whitney *U* tests and exact paired sign-flip permutation tests (2^{-12}). Truth concordance was evaluated with `rtg vcfeval` (`v3.12.1`) against official NIST GIAB and SEQC2 high-confidence BED masks (`HG002 v4.2.1`, `HG002 v5.0q T2T-Q100`, `CMRG v1.00`, `Mosaic v1.10`, `HG005 v4.2.1`, `SEQC2 v1.2.1`, and `HG008 v0.3`/`v0.4`). *In silico* tumor-purity titrations (`20%`, `10%`, `5%`, `2%`, `1%`) were built via seed-controlled `samtools view -s` subsampling and merging of HG008 tumor and normal BAMs (`Supplementary Methods`).

A single-command `VarDictRust` invocation replacing `VarDictJava | teststrandbias.R | var2vcf_valid.pl` is:

```
# Single-binary VarDictRust GERMLINE/SOMATIC calling with native Fisher exact test & VCF emission:
vardictrust -G GRCh38.fa -f 0.01 -N "TUMOR|NORMAL" \
  -b "tumor.bam|normal.bam" -th 32 --fisher -z -c 1 -S 2 -E 3 -g 4 \
  --vcf -A -E-filter 0.05 -P 0 target.bed > calls.vcf
```

Declarations

Ethics approval and consent to participate

Not applicable. All sequencing datasets analyzed in this study are publicly available, open-access reference materials from the NIST Genome in a Bottle (GIAB) Consortium (`HG002`, `HG005`, `HG008`), the FDA SEQC2 Consortium (`HCC1395`/`HCC1395BL`), and the NCBI Sequence Read Archive (`PRJNA744870`/`PRJNA523380`).

Consent for publication

Not applicable.

Availability of data and materials

- **Source Code and Binary Releases:** `VarDictRust` (`v1.0.6`, MIT License) is available at https://github.com/palakpsheth/VarDictRust, with SLSA Level 3 static Linux binaries, Crates.io packages (`vardictrust` and `vardictrust-core`), Bioconda recipes, and `nf-core` modules.
- **Reproducible Benchmark Evidence and Manuscript Compiler:** All `60` benchmark summaries, SHA-256 manifests, CLI scripts, `benchmarks/gcp-study-v106/master_gcp_metrics.json`, `benchmarks/gcp-study-v106/cells_60_inventory.tsv`, `paper/build_figures_and_tables.py`, and `paper/build_manuscript.py` are tracked in the repository. They are archived in `Supplementary Tables S1–S6`.
- **Public Accession Numbers:** Wolff [11] AML panel BAMs: NCBI SRA `SRR15006375`, `SRR15006376`, `SRR15006386`, `SRR15006540` (`PRJNA744870`), and `SRR8657348` (`PRJNA523380`). NIST GIAB reference datasets (`HG002` `v4.2.1`/`v5.0q` `T2T-Q100`/`CMRG v1.00`/`Mosaic v1.10`, `HG005`, and `HG008` `v0.3`/`v0.4` somatic callsets): `https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/` (`HG008` somatic: `data_somatic/HG008/`; draft small-variant benchmarks `Liss_lab/analysis/NIST_HG008-T_somatic-smvar_DraftBenchmark_V0.3-20260425/` and `..._V0.4-20261006/`). SEQC2 `HCC1395`/`HCC1395BL` (`v1.2.1`): `PRJNA506071`.

Competing interests

The author declares no competing financial or non-financial interests.

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Authors' contributions (CRediT)

Palak Sheth: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration. AI pair-programming assistance (`Claude` / `Gemini`) was used under human verification and automated differential oracle-testing gates per the rewrites.bio transparency commitment.

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